Concept Encoding Strategies Across 26 Transformers: A Concept Allocation Zone Evaluation

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Abstract

We evaluate the Concept Allocation Zone (CAZ) framework [Henry, 2026a] across 26 base transformer language models from 8 architecture families (70M–7B parameters). A CAZ is a contiguous window of layers across which a semantic concept assembles into a measurably separable direction in the residual stream — an extended computational event with detectable boundaries and a peak, not a single-layer location — identified from the Fisher-separation curve across depth. We track seven semantic concepts across each model's depth via Fisher-normalized separation, projection ablation, and activation patching.

The central finding: geometric salience is not functional importance. Fisher separation identifies where concepts are most legible; ablation and patching identify where intervention at that layer is causally active — and the gap between legibility and causal magnitude is cohort-systematic. Single-CAZ zone-level ablation (GEM peak protocol, §6.5) is causally active in every cohort (MHA 0.420, GQA 0.261, Gemma 0.168 mean separation reduction), and activation patching at the CAZ peak produces meaningful recovery (MHA 0.597, GQA 0.579, Gemma 0.461, trimmed; raw means 0.753/0.707/0.461 are inflated by overshoot — §6.4). Neither single-CAZ intervention approaches 1.0 in any cohort — concepts are allocated multimodally across multiple CAZes (§4.2), so complete removal requires multi-CAZ protocols (future work). Within that constraint, MHA+GELU models (GPT-2, Pythia, OPT, Phi-2) are the most peak-sensitive cohort, GQA+SwiGLU models (Qwen, Llama, Mistral) are weaker at peak-ablation but comparable at peak-patching, and Gemma-2 (alternating local/global; n=2, case study) localises its functional allocation not at the Fisher peak but at the final global attention layer (near-complete recovery, Table 13). Architecture is the leading hypothesis for the cohort difference, but era, dataset scale, and training curriculum co-vary and are not separable in this sample (§8.3).

Scope of "causal" claims. Throughout this paper, "causal" is operationalized as ablation-measured separation reduction and patching-measured recovery — both geometric measurements inside the residual stream. Neither establishes that CAZ ablation changes next-token probabilities, task accuracy, or generation behavior. §6 states this scope explicitly; §8.7 discusses the behavioral validation status. A behavioral pilot (logit-difference suppression at CAZ peak vs. matched non-CAZ control layer) is reported in §6.8.

Three findings are robust: (1) a cross-architecture concept ordering tendency (median per-model Kendall's $\tau = 0.54$, z = 11.5, $p < 0.001$; 87% of models positively correlated); (2) layer-specific ablation sensitivity at CAZ peaks — projection ablation at detected CAZ layers produces 3.67× greater concept suppression than non-CAZ layers (Mann–Whitney $p = 6.01 \times 10^{-31}$) across all 26 models and 8 architecture families, with the effect persisting across every score category (93–100% of CAZes exceed non-CAZ baseline), and direction-specific ablation at CAZ peak layers suppresses concept-diagnostic token predictions (random-direction ablation at CAZ peak 0; §6.8) — single-layer peak ablation does not consistently outperform a midpoint control because concept directions are distributed across many layers; zone-level behavioral ablation, analogous to the GEM geometric result, is the appropriate follow-up test (§6.8, §8.7); scored detection recovers an additional 422 "gentle" CAZes (score < 0.05) previously invisible to standard prominence thresholds, doubling the causal footprint without diluting its per-feature strength; (3) multimodal concept allocation, with depth-separated sub-representations (shallow vs deep peaks, within-model cos 0.15–0.43) independently converging across architectures. Of seven framework predictions, none are confirmed outright, four are partially supported (including P5 depth-stratified convergence, pending alignment permutation test), one is indeterminate, and two are not supported — with each failure or partial result generating productive dis-

coveries (§5.8). Seven concept probes explain 2% of organized activation structure; 98% remains unlabeled. All experiments run on commodity hardware (2× NVIDIA L4).

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1. Introduction

A central question in mechanistic interpretability is whether transformer language models develop similar internal representations of meaning, or whether each model's representation is idiosyncratic to its architecture and training data. The Concept Allocation Zone framework [Henry, 2026a] formalizes one approach: rather than asking where a concept "lives," it tracks how a concept *assembles* across the depth of the residual stream.

Paper 1 [Henry, 2026a] introduced the framework and its definitions — separation, coherence, velocity, allocation zones — and reported preliminary results across 34 models and 8 architecture families, including instruct-tuned variants. This paper is a systematic evaluation of the framework's predictions by the same research group. We isolate the base models and stress-test the framework's predictions across 26 base models from those same 8 families, characterize phenomena that the framework did not anticipate (multimodal allocation, encoding strategy divergence, gentle CAZes), and quantify the limits of our seven-concept probe set.

A central finding of this evaluation: geometric salience — where a concept is most organized in activation space — does not predict functional importance. Fisher separation identifies the most legible layers; ablation and patching identify the most causally relevant ones. These often diverge, and the divergence is architecturally systematic: MHA models treat geometric peaks as causal bottlenecks; GQA models do not. The practical consequence is that geometric prominence, which the field has used by necessity as a proxy for functional importance, is the right tool for identifying candidate layers but the wrong tool for selecting among them.

1.1 Contributions

- A cross-architecture concept allocation study spanning 26 base models (35 including instruct-tuned variants), 8 architecture families, 623 detected CAZes, and 182 ablation sweeps
- A statistically robust concept ordering tendency across architectures (median per-model $\tau = 0.54$; $z = 11.5$, $p < 0.001$; 87% of models positively correlated), with synthetic training data as the sole genuine inversion
- A scored detection method that recovers 2.6× more structure than threshold detection. A permutation null model confirms that peak separation is 2–10× above noise ($p < 0.01$, 35/35 real concepts; sham concept non-significant, 0/5) while peak count is not diagnostic — ablation in §6.1 independently confirms the newly visible structure is suppression-sensitive (ablation-measured reduction, not behavioral impact)
- A descriptive cohort difference between 2019–2022 MHA+GELU and 2023–2025 GQA+SwiGLU model cohorts in encoding strategy, with GQA+SwiGLU as the leading architectural correlate and training data scale as an undetermined alternative (§8.3)
- A prediction scorecard evaluating all 7 framework predictions: 0 confirmed outright, 4 partially supported (P1, P2, P3, P5 — P5: general convergence confirmed and concept-specific, depth-stratification not supported), 1 indeterminate (P7, underpowered scale test), and 2 not supported (P4, P6) — with each failure or partial result generating productive new directions (§5.8)
- Honest accounting of what we do not explain: 98% of organized activation structure remains unlabeled by our probes

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2. Methods

2.1 Models

We evaluate 26 base transformer language models from 8 architecture families:

Family	Lab	Models	Scale Range	Layers	Position Enc.	Attention	Activation
Pythia	EleutherAI	7	70M–6.9B	6–32	Rotary	MHA	GELU
GPT-2	OpenAI	4	124M–1.5B	12–48	Learned	MHA	GELU
OPT	Meta	5	125M–6.7B	12–32	Learned	MHA	ReLU
Qwen 2.5	Alibaba	4	0.5B–7B	25–29	Rotary	GQA	SwiGLU
Gemma 2	Google	2	2B–9B	26–42	Rotary	GQA	SwiGLU
Llama 3.2	Meta	2	1B–3B	17–29	Rotary	GQA	SwiGLU
Mistral	Mistral AI	1	7B	33	Rotary	GQA+SW	SwiGLU
Phi	Microsoft	1	2.8B	33	Rotary	MHA	GELU

Table 1: Model inventory. MHA = multi-head attention. GQA = grouped-query attention. SW = sliding window. All models run in bfloat16 on 2× NVIDIA L4 (22 GiB each); Gemma-2-9b fits via device-mapped sharding across both GPUs. Architecture papers: Pythia [Biderman et al., 2023], GPT-2 [Radford et al., 2019], OPT [Zhang et al., 2022], Qwen 2.5 [Qwen Team, 2025], Gemma 2 [Gemma Team, 2024], Llama 3.2 [Meta, 2024], Mistral [Jiang et al., 2023], Phi-2 [Javaheripi et al., 2023].

These span learned position embeddings (GPT-2, OPT), rotary (all others); multi-head and grouped-query attention; sliding window attention (Mistral); SwiGLU and GELU activations; and training corpora ranging from The Pile to proprietary multilingual data to synthetic textbooks (Phi-2).

Nine instruct-tuned variants (4 Qwen 2.5, 2 Llama-3.2, 1 Mistral-7B, 2 Gemma-2) were also extracted and are profiled in Supplementary §B; analysis of alignment training effects is reserved for a companion paper [Henry, 2026d].

2.2 Concepts and Datasets

Seven semantic concepts spanning four categories:

Category	Concepts
Epistemic	credibility, certainty
Syntactic	negation
Relational	causation, temporal ordering
Affective	sentiment, moral valence

Each concept is grounded in an established NLP research literature: negation [Morante & Blanco, 2012]; sentiment [Socher et al., 2013; Conneau & Kiela, 2018]; certainty and epistemic modality [Szarvas et al., 2012]; causation [Mirza & Tonelli, 2016]; temporal ordering [Pustejovsky et al., 2003; UzZaman et al., 2013]; moral valence [Haidt & Joseph, 2004]; credibility in the sense of source and claim evaluation [Habernal & Gurevych, 2016]. These concepts were selected for their definitional clarity, contrastive operationalizability, and cross-domain relevance — not for their relation to any specific downstream application.

Each concept is operationalized via 100 contrastive text pairs (positive/negative examples), 150–300 tokens each, balanced for domain (news, academic, conversation, narrative) and difficulty. Pairs were generated by Claude Sonnet 4.5.

Dataset quality: We validated pair quality via a cross-architecture survival test: each pair was scored against a 6-model gauntlet spanning both MHA (Pythia-1.4b, GPT-2-medium, OPT-1.3b) and GQA architectures (Qwen2.5-1.5B, Llama-3.2-3B, Phi-2). A pair "survives" a model if its peak Fisher-normalized separation across layers exceeds 0.3. All 100 pairs per concept survive all 6 models — 100% survival rate across all 7 concepts. Population-level separations range from 0.55 (temporal_order, Llama) to 1.56 (credibility, Phi-2), well above the threshold. The pairs are robustly concept-diagnostic across architecturally diverse models; the results in this paper are not artifacts of model-specific pair tuning.

Dataset provenance and the single-generator concern. All 700 contrastive pairs used in the main study were generated by a single model (Claude Sonnet 4.5). The cross-architecture survival test above demonstrates that every pair produces measurable separation, but it does not demonstrate that the separation is driven by the intended semantic concept rather than by a surface-level generator signature. Single-LLM generation introduces three specific risks: **lexical confounding** (pairs for "credibility" may overuse diagnostic vocabulary like *expert* or *anonymous*), **register gradients** (abstract concepts may be generated with more formal language than concrete ones), and **template convergence** (a single model with a single prompt converges on a limited set of narrative structures, reducing effective diversity below the nominal 100 pairs). These risks propagate silently through the survival test, which checks separation magnitude rather than concept fidelity. We do not treat this as a discharged concern, but it is the concern that most of our validation work targets.

Two observations from the main study partially constrain the artifact concern at the ordering level. First, the concept ordering varies across models: median per-model $\tau = 0.54$, not ~1.0. If the ordering were determined solely by the lexical properties of the pairs, all models receiving the same pairs should produce the same ordering. They do not — Phi-2 inverts the ordering entirely (§3.2), and two additional models show negative correlations. The ordering is an interaction between pairs and models, not a property of the pairs alone. Second, credibility — the concept most vulnerable to lexical confounding (§3.3) — shows the highest cross-model variance (std 29.7), consistent with models resolving the same pairs via different mechanisms rather than all latching onto the same surface signal.

Multi-generator consensus replication (completed, Supplementary §C). The most direct test of the single-generator concern is replication with an independently constructed dataset, and this has been run. A consensus corpus of approximately 1,400 pairs per concept — 100 topics each generated **independently by 14 models spanning four AI labs (Anthropic, Google, OpenAI, Mistral)** — was used to rerun extraction on four architecturally diverse models (Pythia-1.4B, GPT-2-XL, Qwen2.5-3B, Llama-3.2-3B). No model saw a pair from only one generator; the consensus corpus is approximately 14× larger than the main dataset and spans the full generator diversity available to us. The core ordering finding holds: credibility and certainty remain the deepest concepts, negation remains the shallowest, and five of seven concepts sit within $|\Delta| \leq 5$ pp of their single-generator depth across both architectures. The generator-dependent shifts on negation and credibility are diagnostic rather than refutational — they track the multimodal structure independently documented in §3.3 and §4.2 (embedding-level leakage of lexical negation markers in GQA tokenizers; credibility assembled from generator-variable primitives). A single-generator artifact would be expected to produce ordering scrambling; what we observe is ordering preservation with concept-specific depth refinements consistent with the framework's own multimodal predictions.

The remaining planned mitigations — (a) bag-of-words probe baseline to quantify surface-lexical contribution at each depth, (b) pair-level lexical overlap analysis, and (c) human validation of a random 20% sample for concept fidelity — are pair-level audits rather than ordering-level tests; their absence does not reopen the primary concern that the ordering effect is a single-generator artifact. Until (a)–(c) are complete, pair-level lexical confounding could still inflate absolute separation magnitudes, but the ordering result reported in §3 has survived the stronger of the two available tests.

2.3 Metrics

We use three layer-wise metrics defined in [Henry, 2026a]:

Separation $S_c(\ell)$: Fisher-normalized [Fisher, 1936] distance between concept-positive and concept-negative activation centroids in the residual stream, computed as $S_c(\ell) = (\mu_\ell^+ - \mu_\ell^-)^\top \Sigma_\ell^{-1} (\mu_\ell^+ - \mu_\ell^-)/d$. Fisher nor-

malization is critical: it accounts for varying activation scale across layers and models, enabling direct comparison.

Coherence $C_c(\ell)$: Fraction of concept-related variance captured by the top principal component. High coherence = concentrated, low-dimensional representation.

Velocity $v_c(\ell) = dS_c/d\ell$: Layer-wise rate of separation change, smoothed with an adaptive window $k = \max(1, \lfloor L/24 \rfloor)$ where L is the model's layer count. The constant 24 is calibrated to GPT-2-medium (the initial development model), giving $k = 1$ at that depth and scaling proportionally for deeper architectures; it has not been formally validated against ground-truth concept boundaries.

Adjacent-layer CKA $\mathrm{CKA}(\ell, \ell{+}1)$: Linear Centered Kernel Alignment [Kornblith et al., 2019] between the residual stream at consecutive layers. For layer activations $X \in \mathbb{R}^{n \times d}$, the linear kernel is $K = XX^\top$, and $\mathrm{CKA}(K_\ell, K_{\ell+1}) = \mathrm{HSIC}(K_\ell, K_{\ell+1}) / \sqrt{\mathrm{HSIC}(K_\ell, K_\ell) \cdot \mathrm{HSIC}(K_{\ell+1}, K_{\ell+1})}$ using the unbiased HSIC estimator. Values near 1 indicate representational stability (little transformation between layers); dips mark active transformation windows. Used in §6.7 to refine CAZ extent and characterize inter-CAZ coasting regions.

2.4 Detection Methods

Legacy (threshold): CAZ boundaries defined by velocity zero-crossings with a 10% prominence floor. Detects 201 CAZes across 22 models.

Scored (primary): A composite score incorporating three features:

$$\text{CAZ score} = \text{prominence} \times \text{coherence boost} \times \sqrt{\text{width}}$$

with a 0.5% prominence floor. This detects 623 CAZes across 26 models — a 3.1× increase. We categorize CAZes by score:

Category	Score Range	Count	% of Total
Black hole	> 0.5	103	17%
Strong	0.2–0.5	86	14%
Moderate	0.05–0.2	117	19%
Gentle	< 0.05	317	51%

Table 2: Score distribution of 623 detected CAZes across 26 models. Updated from 201/22 models under the legacy threshold detector.

2.5 Causal Testing: Ablation and Activation Patching

We use two complementary causal tests at each identified CAZ peak. Activation patching here is not a novel method but a downstream validation step applied to CAZ-identified layers: the framework paper [Henry 2026a §5.1] positions CAZ detection as upstream of the activation-patching toolkit [Vig et al. 2020; Meng et al. 2022; Wang et al. 2023; Chan et al. 2022; Goldowsky-Dill et al. 2023], with the velocity-based and score-based region detectors supplying a principled choice of *which* layers to test rather than exhaustive layer sweeps. The results in §6 are therefore best read as causal validation of a geometric hypothesis, not as an independent patching study.

Projection ablation: Orthogonal projection removes the concept's dominant direction from the layer output. Two measurements per ablation: separation reduction (concept suppression, 50 contrastive pairs) and KL divergence from the unablated distribution (capability damage, 12 general-capability prompts). The ratio identifies the optimal intervention point.

Activation patching: Mean-field shift patching [Meng et al. 2022; Geiger et al. 2021; cf. Vig et al. 2020 for the broader causal-mediation framing] injects the concept at a layer from the opposite direction — instead of removing the signal, it adds it. For each layer L, we compute the mean activation shift between positive

and negative classes: $\Delta_L = \mu_L^+ - \mu_L^-$, add this shift to all negative hidden states at L, and run the remaining forward pass normally. Performance is measured via **concept score recovery**:

$$\text{recovery}(L) = \frac{\bar{s}_{\text{patched-neg}}(L) - \bar{s}_{\text{neg}}}{\bar{s}_{\text{pos}} - \bar{s}_{\text{neg}}}$$

where scores are projections onto the final-layer concept direction. Recovery of 0 means the layer is causally inert for injection; 1.0 means patching fully restores concept encoding at the output; >1.0 indicates overshoot (patched negatives exceed the positive class centroid). Ablation and patching together distinguish *ablation-sensitive* layers (ablation suppresses measured separation; patching restores measured separation) from *injection-responsive* layers (patching restores measured separation; ablation has no effect) — a dissociation that turns out to be architecturally informative (Section 6.4). We use "necessary" and "sufficient" as informal shorthand for these roles in the surrounding text; both metrics fall short of 1.0, indicating partial redundancy even in MHA models.

2.6 Hardware and Precision

All experiments were conducted on 2× NVIDIA L4 GPUs (22 GiB each). Forward passes in bfloat16 for all models, including Gemma-2-9b (fit via device-mapped sharding across both L4s; ~18 GB of weights at bf16). All Fisher separation, coherence, and Procrustes computations in float64 — this corrects a silent overflow discovered in fp16 Fisher normalization at deep layers of models with >36 layers.

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3. Concept Ordering

3.1 Cross-Family Ordering

Averaging CAZ peak depth across all 26 base models:

Rank	Concept	Mean Depth (%)	Std	Category
1 (earliest)	credibility	39.6	29.7	epistemic
2	negation	49.8	17.9	syntactic
3	causation	53.6	16.6	relational
4	temporal_order	56.1	17.1	relational
5	sentiment	61.8	13.6	affective
6	certainty	64.2	14.2	epistemic
7 (deepest)	moral_valence	68.1	14.1	affective

Table 3: Mean CAZ peak depth across 26 base models from 8 architecture families. Values reflect the dominant (tallest) peak per concept per model. For multimodal concepts (Section 4.2), this is whichever of multiple peaks happened to be tallest — contributing to inflated variance for credibility (std 29.7) where peak-switching between shallow and deep sub-representations accounts for much of the spread.

The category-level ordering — syntactic/epistemic earliest, relational mid-stream, affective deepest — holds across 7 of 8 families. Within categories, the ordering is stable: causation precedes temporal ordering; sentiment precedes moral valence.

Figure 1. CAZ peak depth (% of model depth) across all 26 base models and 7 concepts. Rows are concepts ordered by mean peak depth (shallowest at top); columns are models grouped by architecture family and sorted by scale within each family. Green = shallow (early layers), red = deep (late layers). The concept ordering tendency — credibility and negation shallow, moral valence and certainty deep — is visible as a cool-to-warm gradient from top to bottom that persists across all architecture families. Phi-2 (synthetic textbook training) is the sole visible inversion (§3.2).

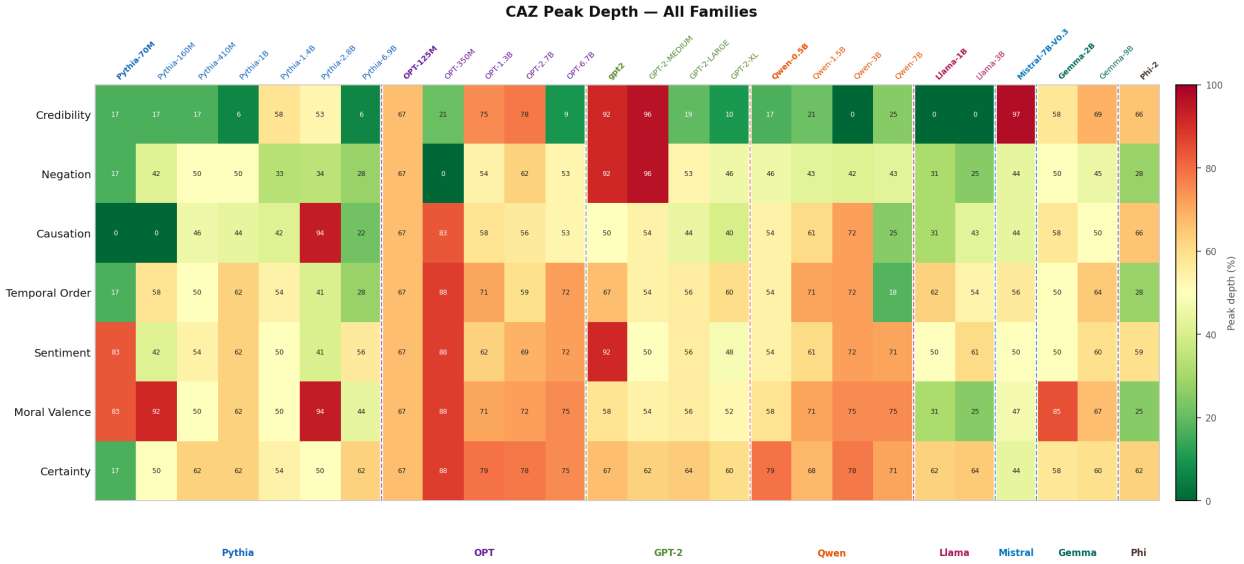

Figure 1: Peak depth heatmap across 26 models × 7 concepts

To quantify cross-model consistency, we compute Kendall's τ between each model's concept depth ranking and the mean ordering (Table 3). Of 24 models with non-degenerate rankings (excluding OPT-125m, which shows a floor effect with all concepts at identical depth), 21 (87.5%) have positive τ, and 11 (45.8%) are individually significant at $p < 0.05$. Three models show negative τ: Phi-2 (synthetic training, Section 3.2), GPT-2, and GPT-2-medium (both small-scale models with limited depth differentiation). The mean τ vs reference is 0.45 (median 0.54). Pairwise Kendall's τ across all 276 model pairs gives a mean of 0.24, with 71.7% of pairs concordant. A permutation test ($n = 5000$, shuffling concept depths independently within each model) yields $z = 11.5$, $p < 0.001$ — the ordering is statistically significant, not an artifact of averaging.

Stability varies by concept. Sentiment (std 13.6), moral valence (std 14.1), and certainty (std 14.2) are the most consistent across architectures. Credibility (std 29.7) is the least stable — a bimodality we examine in Section 3.3.

Figure 2. *Negation* CAZ profiles for all 26 base models, grouped by architecture family and sorted by scale. Fisher separation $S(\ell)$ (coloured curve), CKA-refined CAZ bands (narrow shaded region), and adjacent-layer CKA similarity (dashed, right axis) per panel. Negation is the second-shallowest concept (mean 49.8%) with the second-lowest variance (std 17.9%), making it a clean illustration of cross-family ordering stability. The consistent peak position near 40–55% depth — visible across all 8 families — provides the within-concept view that the heatmap (Figure 1) compresses.

3.2 The Phi-2 Inversion

Interpretive caution: Phi-2 is a single model with a non-standard training process (synthetic data, knowledge distillation). With n=1, the effects of synthetic data, distillation, and curriculum cannot be cleanly separated. The inversion is reported as an observation, not explained.

Phi-2 (Microsoft, 2.8B parameters, MHA+GELU) is the most extreme outlier in concept ordering ($\tau = -0.25$ vs reference). Its peak depths:

Concept	Phi-2 Depth (%)	Mean Rank
moral_valence	25.8	7 (deepest)
negation	29.0	2
temporal_order	29.0	4
sentiment	61.3	5

Concept	Phi-2 Depth (%)	Mean Rank
certainty	64.5	6
credibility	67.7	1 (earliest)
causation	67.7	3

Affective and syntactic concepts assemble first (25–29%), epistemic and relational concepts last (62–68%) — the opposite of every other model. This inversion is consistent across all 7 concepts, ruling out measurement noise. Coherence at the early affective peaks is moderate (0.11–0.18), lower than the deep epistemic peaks (0.25–0.34). The early assembly is real but geometrically less consolidated — whatever drives the inversion produces earlier affective peaks with lower coherence than the deep epistemic peaks in models that follow the standard ordering. Phi-2 is the only model in our sample trained primarily on synthetic data (textbooks generated by GPT-3.5), and it was also trained with knowledge distillation from a larger model — a different training *process*, not just different data. Its architecture (33 layers, 2560 hidden dim, MHA, rotary, GELU) closely matches Pythia-2.8b (32 layers, 2560 hidden dim, same attention and activation), which follows the concept ordering. This makes architecture an unlikely explanation. One strong inversion does not refute P2, but it constrains the claim: the ordering tendency may be a property of models trained on natural text, not of transformer architectures in general. The inversion also serves as a partial control on dataset artifact: Phi-2 receives the same contrastive pairs as every other model but produces the opposite ordering. If the ordering were determined solely by the lexical properties of the LLM-generated pairs (e.g., a concreteness gradient from credibility to moral valence), Phi-2 should show the same ordering as all other models. It does not, indicating that the ordering is at minimum an interaction between dataset and model, not a property of the dataset alone (see §2.2 for further discussion).

Overall: 21/24 non-degenerate base models show positive rank correlation with the mean concept ordering. The consistency is statistically significant (z = 11.5) but varies from near-perfect ($\tau = 0.93$, OPT-6.7B) to weakly negative ($\tau = -0.37$, GPT-2-medium). The ordering is a robust statistical tendency, not a strict per-model invariant.

3.3 The Credibility Puzzle

Credibility exhibits the highest cross-model variance (std 29.7) despite having the strongest Fisher separation of any concept (mean $S = 1.16$). Three distinct behaviors emerge:

- **Embedding leakage** (12/26 models): Peaks below 25% depth. Models resolve credibility from surface lexical cues near the embedding layer.
- **Late assembly** (5/26 models, including Mistral-7B at 100%): Peaks above 75% depth. Genuine multi-layer compositional assembly.
- **Mid-stream assembly** (9/26 models): Peaks at 25–75% depth, consistent with CAZ prediction.

The embedding-leakage count is partly generator-dependent: the multi-generator consensus replication (§C) shows that credibility depth shifts across generators track this leakage structure, consistent with the primitives underlying shallow credibility resolution varying across generation sources. The late-assembly and mid-stream counts are stable across generators.

Within the GPT-2 family, credibility inverts with scale: small models peak at 91–96% depth (deep inference required), while large models peak at 10–19% (resolved from embeddings). Llama-3.2 base models show embedding leakage at both scales (0% depth), while Llama-3.2-1B-Instruct shifts credibility to 80% — instruct training moves it from surface cues to deep computation.

Across 17 models with verified multimodal credibility (shallow and deep peaks well-separated in depth), the two peak directions are geometrically distinct: within-model cosine similarity between shallow and deep peak directions averages 0.379 (range 0.14–0.59; only 4 of 17 exceed cos = 0.5; $p < 0.001$). Models converge separately on a shallow lexical credibility representation and a deep compositional one. The *what* of credibility representation is universal; the *when* is architecture-dependent.

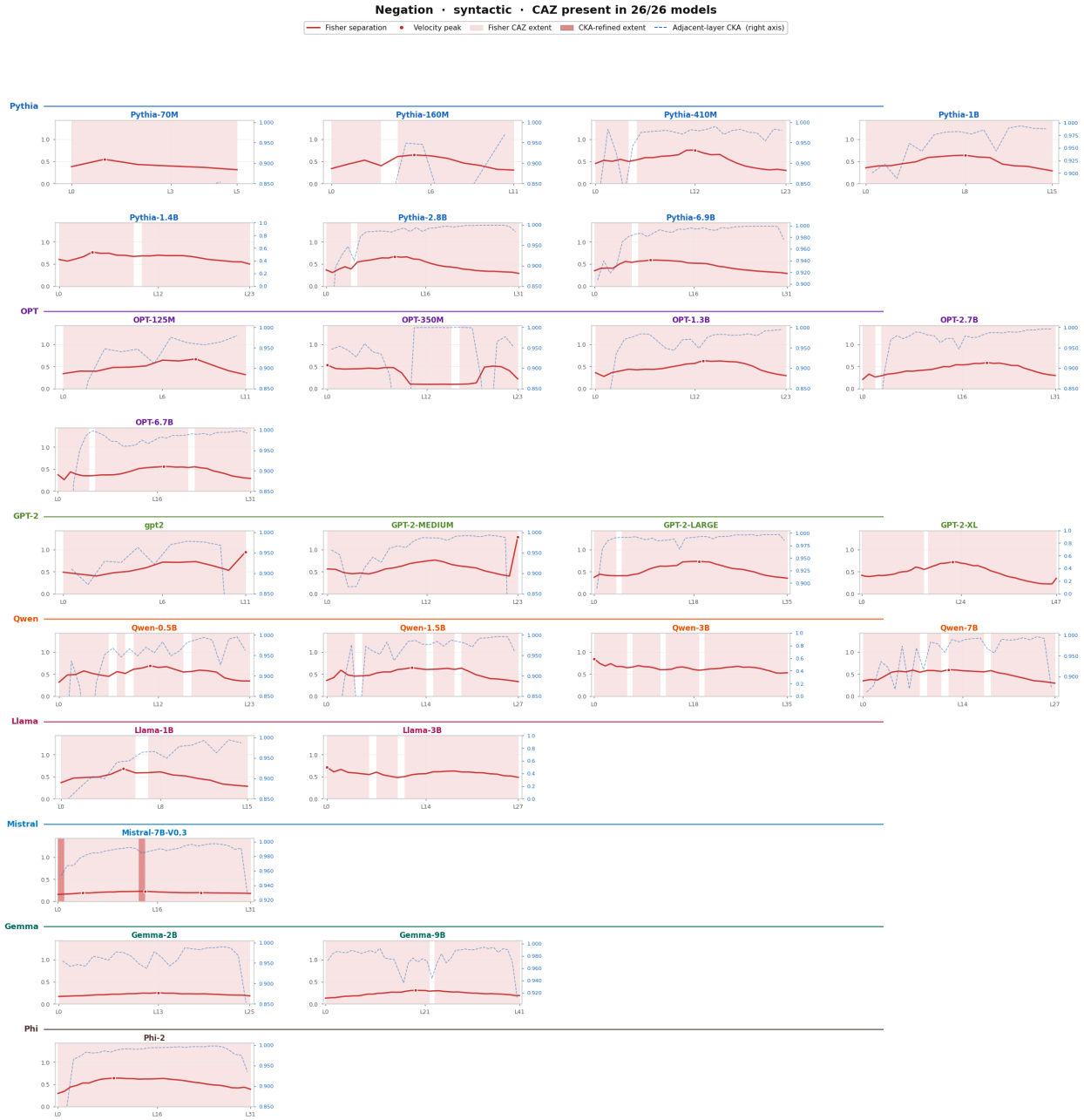

Figure 2: Negation CAZ profiles across all 26 base models

4. Scored Detection and CAZ Profiles

4.1 The Detection Gap

Threshold-based detection (10% prominence floor) identifies 201 CAZes. Scored detection (0.5% prominence floor, composite scoring) identifies 623. The 422 additional CAZes are predominantly gentle (score < 0.05), distributed more uniformly across depth than strong CAZes.

Permutation null model. A natural concern is that lowering the prominence threshold simply captures noise fluctuations in the Fisher separation trace. We tested this directly: for each of 5 representative models (Pythia-70m, Pythia-1.4b, GPT-2-XL, Qwen-2.5-3B, Gemma-2-2b) $\times$ 7 concepts, we randomly permuted pos/neg labels within each contrastive pair 100 times and re-ran the full detection pipeline. As a sanity check, we included a sham "concept" — 100 pairs of randomly assigned texts with no semantic contrast.

The null model produces a comparable number of velocity peaks: approximately 1 per 6–12 layers regardless of whether labels carry concept signal (mean null peaks: 1.2 at 6 layers, 2.7 at 24, 6.4 at 36, 4.8 at 48; per-layer density ranges from 1/5 at 6 layers to 1/10 at 48 layers). Peak count is not diagnostic of concept structure — it is a property of the detection pipeline's sensitivity to fluctuations in any non-monotonic trace. No real concept achieved $p < 0.05$ on peak count (0/35).

What distinguishes real concepts from noise is the separation magnitude at detected peaks. Across all 35 real model $\times$ concept combinations, observed Fisher separation at the peak was 2–10$\times$ above the permutation null ($p < 0.01$ in all 35 cases; MHA models: 4–10$\times$; alternating-attention: 2–3$\times$). The sham concept was indistinguishable from noise on all 5 models (observed/null ratio 0.7–1.0$\times$; $p = 0.34$–0.78). The pipeline correctly classified 40/40 combinations: every real concept significant, every sham non-significant.

	N	Obs/Null sep ratio	p(sep) < 0.01
Real concepts	35	2–10$\times$	35/35 (100%)
Sham concept	5	0.7–1.0$\times$	0/5 (0%)

Table 4: Permutation null model results across 5 architectures. 100 permutations per combination. Sham = 100 randomly paired texts with no semantic contrast.

The reported 623 CAZes and 3.4-per-concept mean should be understood in this context: the count is expected under noise; the signal is not. What makes these peaks meaningful is not their number but their separation magnitude (confirmed here) and their causal efficacy (confirmed by ablation in §6.1).

Peak CAZ activity occurs at 50–60% depth (80 CAZes, mean score 0.339). The 30–40% range is the quietest (62 CAZes, mean score 0.099), creating a relative void between early processing and main allocation. This depth distribution may partially reflect a generic property of transformers — middle layers are routinely identified as the locus of abstract processing in unrelated work — rather than a concept-allocation-specific effect.

A note on score interpretation across architectures. CAZ scores reflect both semantic geometry and architectural amplification. In MHA architectures, where attention at every layer reinforces the concept direction, separation peaks are amplified by the reinforcement loop — producing high scores that correlate with genuine ablation impact. In GQA and alternating-attention architectures, this amplification is absent or attenuated: the same semantic function registers as a lower (or locally identical) score. Gemma-2, for example, produces scores in the "black hole" range (0.26–0.60) at its identified peaks yet shows zero separation reduction when those peaks are ablated. The score categories (black hole, strong, moderate, gentle) are meaningful within an architecture family; cross-family comparisons require the architecture-conditioned analysis in Section 6.4. The aggregate count of 623 CAZes is based on velocity zero-crossing detection, which is architecture-independent; the score categories are descriptive labels applied post-detection and should not be read as functionally equivalent across architecture families.

4.2 Multimodal Allocation

Concepts do not assemble at a single layer. Across 26 models and 7 concepts, we observe a mean of 3.4 CAZes per concept per model. Credibility is the most multimodal (73% of models show 2+ peaks), but every concept shows multi-modality in at least some models. As shown in §4.1, the peak count itself is not diagnostic — the permutation null produces a comparable number of velocity peaks in any non-monotonic trace. What makes these peaks meaningful is the separation magnitude at each peak (2–10× above null, §4.1) and the causal efficacy of ablation at those peaks (3.67× greater suppression than at non-CAZ layers, §6.1).

The sub-representations at different depths are geometrically distinct:

Concept	N multimodal	Mean cos(shallow, deep)
credibility	17	0.379
sentiment	2	0.433
certainty	6	0.294
causation	6	0.212
moral_valence	4	0.241
negation	6	0.206
temporal_order	5	0.156

Table 5: Cosine similarity between shallow and deep peak directions within the same model. Values of 0.15–0.43 are well above random baseline (|cos| ≈ 0.02 for random unit vectors in these dimensions) but well below identity — consistent with either geometrically distinct sub-representations or a single diffuse concept direction that rotates gradually across depth.

The farther apart the peaks, the more different the directions: $r = -0.496$, $p = 0.0005$. This correlation is consistent with both interpretations — genuine sub-representations at different depths, or natural rotation over distance. Distinguishing these requires showing that the transition between peaks is discontinuous rather than smooth; the current data do not resolve this.

We compared the saddle point boundary definition against an independent geometric signal: adjacent-layer direction rotation, measured as $|\cos(d_\ell, d_{\ell+1})|$ where d_ℓ is the dominant concept direction at layer ℓ. A sharp rotation event (local minimum of $|\cos| < 0.7$) marks a layer where the model's top concept direction pivots near-orthogonally. Across all 26 models and 7 concepts, 50% of separation saddle points fall within ± 2 layers of a direction rotation event — indistinguishable from a permuted baseline of 53% (ratio 1.07×; $N = 2000$ permutations shuffling rotation positions within each model).

The two boundary definitions are essentially uncorrelated. This null result is ambiguous: it could mean that saddle points and direction rotations track genuinely distinct phenomena (functional separability changes vs. geometric direction changes), or it could mean that saddle points are not tracking structurally meaningful boundaries at all. We report it without privileging either interpretation.

Multi-modality does not correlate with model scale ($\rho = 0.11$, $p = 0.63$). It varies by architecture family: Qwen 2.5 shows deep bimodality (dips of 26–36% between peaks), Gemma 2 shows subtle bimodality (dips of 8–15%), and GPT-2 and Pythia are intermediate.

The inter-peak saddle does not represent concept absence. Across 895 inter-peak saddle points, separation at the saddle retains a mean of 89% of adjacent peak separation (median 92%), ranging from ~79–82% in MHA families to >95% in alternating-attention models. The concept is geometrically present throughout the inter-peak interval; the saddle marks a region of reduced assembly velocity, not a gap in concept representation.

4.3 Depth-Dependent Concept-Pair Geometry

After Procrustes alignment into a shared coordinate space, concept pairs show consistent depth-dependent relationships. These patterns hold across all 20 same-dimension model pairs tested (5 architecture families):

Concept Pair	Category	Shallow alignment	Deep alignment	Direction	Models with pattern
causation × temporal_order	relational	moderate	higher	converge	20/20
credibility × certainty	epistemic	moderate	lower	diverge	20/20
sentiment × moral_valence	affective	moderate	lower	diverge	18/20

Table 6: Depth-dependent concept-pair geometry across 20 same-dimension model pairs. Cosine values are omitted because the Procrustes alignment is heavily underdetermined (see §4.3 caveat); only the direction of the depth-dependent change is reported. Do not cite these as measured quantities pending the permutation-based validation in [Henry, 2026c].

A methodological caveat on Procrustes alignment. Aligning 7 concept directions in d-dimensional space ($d \gg 7$) leaves the rotation matrix heavily underdetermined — effectively 21 constraints for $d(d-1)/2$ degrees of freedom. The absolute cosine values in Table 6 may be inflated by alignment overfitting, and the sign of small deltas is not guaranteed to survive that inflation either: a poorly-constrained rotation can in principle flip the sign of modest effects along with their magnitude. What is reported here is that the direction of the delta (converge vs. diverge) is *consistent* across all 20 model pairs — a descriptive pattern, not a validated directional claim. Full analysis of Procrustes alignment quality, including permutation-based significance tests that address both the magnitude-inflation and sign-robustness concerns, is the subject of a companion paper [Henry, 2026c] and should be consulted before citing the numbers in Table 6 as measured quantities.

A parsimonious alternative deserves equal weight with the allocation-dynamics reading. Causation and temporal ordering are semantically related in any distributional-semantics sense — cause precedes effect, and the training corpus reflects this — so any model that learns temporal structure will represent the two concepts more similarly at layers where temporal reasoning is complete, regardless of architecture-specific allocation mechanisms. Credibility and certainty share surface cues ("I'm certain" ≈ "credible") at shallow depths but functionally dissociate once the model can distinguish speaker confidence from claim support, which is also a depth-dependent property of representation learning on natural text. Under this reading, the converge/diverge pattern is downstream of shared training distributions, not of anything specific to the CAZ framework. All 26 models were trained on overlapping internet-text corpora; the 20/20 consistency establishes that the depth-dependent geometry is *robust*, but not that it is *caused by* concept allocation dynamics rather than by semantic relatedness at different levels of abstraction. Distinguishing these readings would require either (a) models trained on deliberately non-overlapping corpora to see if the pattern dissociates from training data, or (b) a mechanistic intervention (e.g., ablation of a shared shallow direction, §4.5) showing that the converging/diverging concepts causally interact in a way that pure distributional geometry would not predict. Neither has been run. This section reports a pattern; the mechanism behind it is unresolved.

4.4 Family-Level Score Distributions

Scored detection reveals distinct family-level patterns in how the pipeline's metrics distribute across architectures:

- **Gemma 2**: Near-zero black holes (0–1 per model). Many gentle CAZes (21–37 per model, artifacts excluded). Separation changes are distributed across depth with low per-layer amplitude — consistent with the alternating local/global attention mechanism producing smoother separation profiles.
- **Qwen 2.5**: High CAZ counts (34–43) with many gentle CAZes. Broad distribution across depth.
- **GPT-2**: Fewer but stronger peaks. More concentrated separation changes with higher mean scores.
- **Small models** (Pythia-70m, OPT-125m): High CAZ-per-layer density (0.8–1.7). This is partly arithmetic — fewer layers compress the same detection events into a smaller space.

These differences are expected: different architectures produce different score distributions. Whether they reflect distinct functional strategies or are simply measurement artifacts of running the same pipeline on different architectures is not resolved by the score distributions alone. The causal analysis in §6 provides partial disambiguation.

4.5 Cross-Concept Shallow Sharing

Shallow-peak dominant directions reveal shared structure across concepts:

Concept Pair	Mean cos(shallow)	Notable
certainty × credibility	0.456	opt-6.7b: **0.926** — near-identical direction
causation × negation	0.456	
causation × temporal_order	0.363	Relational concepts share primitives
credibility × sentiment	0.074	Unrelated
causation × credibility	0.014	Orthogonal

Table 7: Cross-concept cosine similarity of shallow-peak dominant vectors.

The shallow CAZ for credibility in opt-6.7b is essentially the same feature as the shallow CAZ for certainty (cos 0.926). Cross-category pairs (epistemic × affective, relational × epistemic) are near zero. This is consistent with the standard finding in probing studies that early layers encode generic features while later layers encode task-specific ones. One interpretation is that the unit of analysis at shallow depths is a shared *allocation primitive* — a discrete computational direction that multiple concepts share before diverging at depth. An equally valid interpretation is that early layers simply have not yet separated semantically related concepts: the shared direction reflects incomplete processing, not a functional primitive. Distinguishing these interpretations would require showing that the shared shallow direction is itself a causal unit (e.g., that ablating it disrupts both concepts). That experiment has not been run.

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5. Prediction Scorecard

The CAZ framework [Henry, 2026a] generated testable predictions. We evaluate each against the 26-model dataset.

5.1 P1: Optimal Ablation Depth — PARTIALLY SUPPORTED

Prediction: Ablation within the CAZ produces the best suppression-to-damage ratio.

Result: Across 204 ablation sweeps (26 models × 7 concepts, with multimodal concepts contributing one sweep per detected region), within-CAZ is the modal class for optimal ablation (56.9% within, 32.4% post-CAZ, 10.8% pre-CAZ; median optimal depth 59.5%). The pattern holds across architecture cohorts: MHA 53.1% within (median 60%), GQA 72.1% within (median 56%), Gemma 50.0% within (median 59%). The *location* claim of the prediction is supported at majority level — but the prediction's stronger reading, that the best point coincides with the CAZ peak itself, is not: the optimum consistently drifts to CAZ edges or nearby layers rather than landing at the Fisher-peak layer. P1 is supported as a region-level claim and not supported as a peak-level claim.

The reason the optimum is broad rather than peaked: in many models, concept separation reduction is roughly constant across a band of layers (within-CAZ CV 0.02–0.26 for MHA). The concept direction is redundantly encoded across adjacent layers of the residual stream within the CAZ region, so projection ablation succeeds wherever the concept has been committed — including layers just inside, just outside, or

adjacent to the detected peak. What varies more sharply is collateral damage. The CAZ marks where a concept is most *geometrically prominent*; the optimum suppression-to-damage point is typically nearby but not necessarily *at* that prominence peak.

A separate finding from the residual drift. The peak-vs-region distinction led us to compare two causal protocols directly. Activation patching tells a different story than projection ablation. Patching at the CAZ peak injects a valid concept representation into the residual stream at a point where the network can detect and propagate it. Across all 26 models, patching at or near the CAZ peak produces meaningful recovery (MHA: 0.753; GQA: 0.707; Gemma: 0.461 at peak, ~1.0 at final global layer — see Section 6.4). CAZ peaks are consistently valid *injection sites* whose peak-ablation efficacy ranks lower (MHA 0.420, GQA 0.261 sep-reduction under GEM peak ablation) — single-CAZ removal is causally active but not concept-terminating in any cohort, because concepts are multimodally allocated (§4.2). The distinction matters for applications: ablation-based safety interventions require knowing where the concept cannot be recovered from; patching-based steering requires knowing where the concept can be reliably introduced.

The architectural generation effect: Modern architectures show a different pattern.

Architecture Era	Families	Sep. Reduction CV	Encoding
Legacy (2019–2022)	GPT-2, Pythia, OPT	0.02–0.26	Redundant
Modern (2024–2025)	Qwen 2.5, Gemma 2	4.78–6.40	Sparse

Table 8: Encoding strategy by architecture era. CV = coefficient of variation of separation reduction across layers.

In Qwen and Gemma, concept encoding is layer-specific — ablation works strongly at some layers and barely at others. The cohort effect is on sharpness of the optimum, not on its position: all three cohorts locate their median optimum near mid-depth (MHA 60%, GQA 56%, Gemma 59%), but the MHA optimum is broad (many neighboring layers are near-optimal) while the GQA/Gemma optimum is narrow (mis-targeted ablation fails). Within the 2023–2025 GQA cohort, the sharpest suppression layers lie near the end of the assembly chain (typically the final handoff into the unembedding projection) rather than at the Fisher peak itself — which is one reason peak-ablation is weaker than peak-patching in that cohort (§6.4).

5.2 P2: Architecture-Stable Ordering — PARTIALLY SUPPORTED

Prediction: The relative ordering of concept assembly depths is stable across architectures.

Result: 87% of base models (21/24 non-degenerate) show positive rank correlation with the mean concept ordering (median $\tau = 0.54$). Pairwise consistency across all model pairs is significant ($z = 11.5$, $p < 0.001$). The tendency is real and statistically robust — but $\tau = 0.54$ is moderate rank correlation, not the stable ordering the prediction claimed. Three models show inversions (Phi-2, GPT-2, GPT-2-medium), and the prediction said "stable," not "tends toward." The ordering is a significant cross-architecture tendency, not a strict invariant.

5.3 P3: Width Scales with Abstraction — PARTIALLY SUPPORTED

Prediction: More abstract concepts have wider CAZes.

Result: Excluding credibility (bimodal, high width variance), the correlation between abstraction level and CAZ width is $r = 0.294$, $p = 0.003$ ($n = 132$ sweeps). Pairwise comparisons are significant: certainty wider than negation by 18.5% ($p = 0.002$), moral valence wider than causation by 18.3% ($p = 0.007$). 15 of 17 models with sufficient depth show positive within-model correlations.

Credibility confounds the overall test (not significant with credibility included, $p = 0.186$). Sentiment also deviates — narrow CAZ despite being affective. The effect exists for concepts where assembly is genuinely compositional and breaks for concepts where surface cues dominate.

5.4 P4: Concept Handoffs and Bandwidth Recycling — NOT SUPPORTED

Prediction: Post-CAZ separation loss correlates with proximity to the unembedding layer.

Result: The prediction assumes a single CAZ peak per concept per model. Multimodal allocation (mean 3.4 CAZes per concept per model) invalidates this assumption — apparent "decay" between peaks is not degradation but inter-CAZ gaps where the geometric direction is reallocated. The prediction as stated is not testable against multimodal data.

A productive failure. Reframing decay as post-*chain* degradation — separation loss from the final CAZ peak to the last layer — reveals structure that the original prediction was reaching for but could not formalize. Two patterns emerge from this reframing:

Velocity X-crossings. Plotting velocity curves for multiple concepts on the same axis reveals a pattern: in Pythia-6.9b, $v_{\text{causation}}$ goes negative at L9 (29% depth) within one layer of where $v_{\text{certainty}}$ goes positive — a velocity X-crossing. A second X-crossing occurs near L20 (64% depth), where $v_{\text{certainty}}$ goes negative as $v_{\text{credibility}}$ goes positive. Figure 3 shows the corresponding pattern in GPT-2-XL; the Pythia-6.9b profile is not plotted separately but follows the same qualitative geometry at different absolute layer positions. These crossings align with the relay transitions identified in §7 (causation → certainty → credibility). One interpretation is that post-CAZ degradation reflects geometric bandwidth recycling, where Concept A's direction is consumed to build Concept B. This is consistent with the data but not demonstrated by it — the correlation between velocity crossings and relay transitions could also reflect shared depth-dependent structure rather than a causal link.

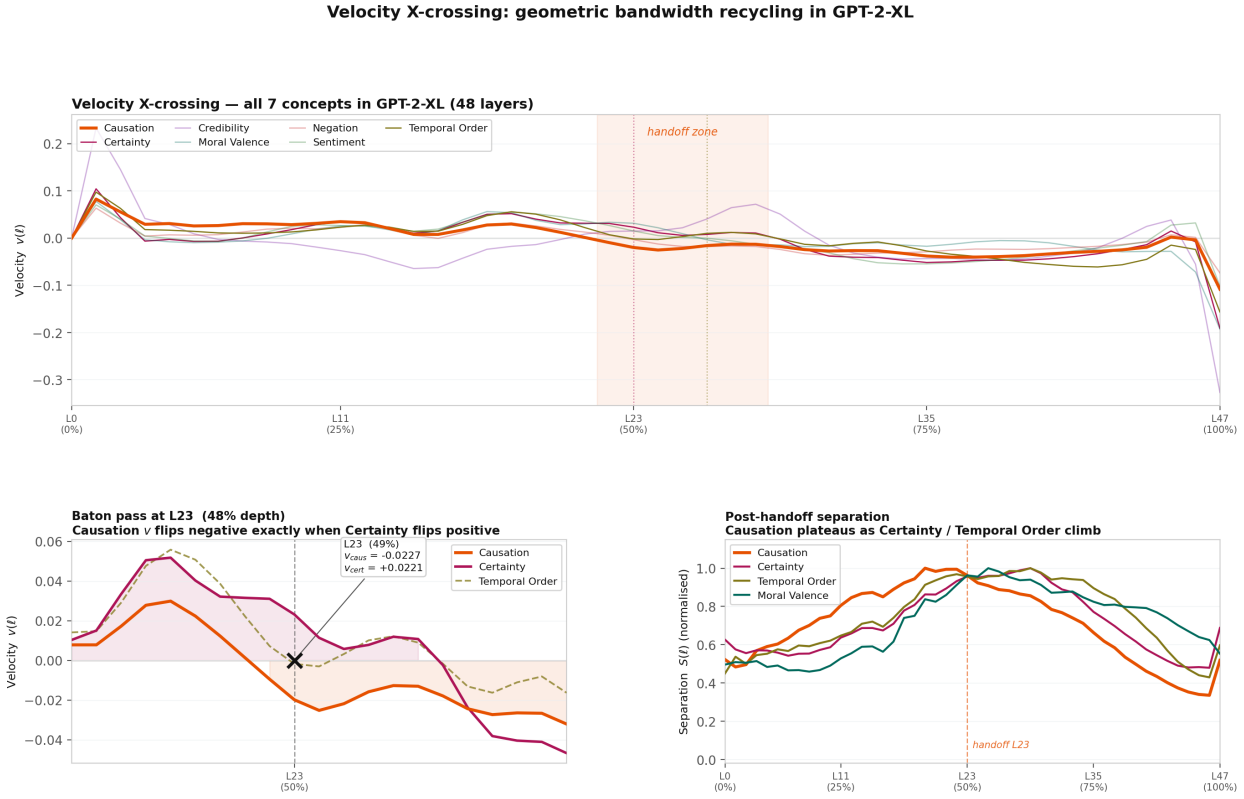

Figure 3: Velocity X-crossing baton pass in GPT-2-XL

Figure 3. Velocity X-crossing in GPT-2-XL (48 layers). **Top:** All 7 concept velocity curves; causation highlighted. The handoff zone (L22–L28, shaded) marks where causation velocity goes negative. **Bottom-left:** Close-up of the causation × certainty × temporal order crossing at L23 (48% depth). The large × marks the precise layer where $v_{\text{causation}}$ flips negative and $v_{\text{certainty}}$ flips positive simultaneously — the

mathematical signature of bandwidth recycling. **Bottom-right:** Separation curves (normalised) showing causation plateauing at the handoff layer while certainty and temporal order continue to climb. The Pythia-6.9b profile shows the same qualitative structure at L9 and L20 (not plotted; layer-level velocity data available in the supplementary extraction outputs).

Results (182 measurements, 26 models):

(a) Remaining depth predicts decay: $r = -0.312$, $p < 0.001$. More layers between the final CAZ and the output = more separation loss. Mean decay ratio 0.725 (concepts retain ~73% of peak separation through to the final layer).

(b) Unembedding token clustering: $r = 0.173$, $p = 0.019$. Concepts whose tokens cluster more tightly in unembedding space decay less. Significant but weak — proximity to the output layer is the dominant factor; unembedding structure contributes modestly.

Concept	Mean Final Peak (%)	Mean Decay Ratio	Mean Remaining Depth (%)
credibility	79.0	0.915	20.0
sentiment	71.9	0.848	26.6
moral_valence	75.3	0.694	23.7
causation	61.4	0.669	36.5
certainty	70.5	0.666	28.0
negation	64.2	0.643	33.9
temporal_order	72.5	0.640	26.2

Table 9: Post-chain degradation by concept. Concepts with later final peaks and less remaining depth show less decay.

Both boundaries of the transformer constrain what we measure: embedding-layer CAZes reflect tokenizer features rather than transformer computation, and post-chain decay is shaped by proximity to the unembedding projection. A dedicated analysis of how token-space structure at both ends constrains concept geometry in the intervening layers remains future work.

5.5 P5: Depth-Stratified Convergence — PARTIALLY SUPPORTED (general convergence confirmed; depth-stratification not supported)

Prediction: Multimodal concepts show depth-matched cross-architecture alignment.

Result: The prediction decomposes into three sub-claims, each tested by a separate phase of the alignment validation study [Henry, 2026c]:

Sub-claim 1 — General cross-architecture convergence: CONFIRMED. Phase 7a/7c establish that cross-architecture alignment is genuine, not a Procrustes artifact. Grand mean aligned cosine: 0.9807 (raw unaligned baseline 0). The permutation-label null test (Phase 7c; 100 trials per concept, labels shuffled before Procrustes fit) yields per-concept null means of 0.003–0.006 — z-scores of 22–37 per concept, all $p < 10^{-115}$. The alignment is not overfitting to the underdetermined rotation.

Sub-claim 2 — Rotation is concept-specific: CONFIRMED. Phase 7d shows that the Procrustes rotation learned on one concept's directions does not transfer to another concept. Same-concept transfer: mean cosine 0.9807; cross-concept transfer: mean cosine 0.1904 ($t = 82.35$, $p \approx 0$). Per-concept same-to-cross ratios range from 4.2× (causation) to 9.4× (credibility). The rotation encodes concept identity, not generic representational geometry.

Sub-claim 3 — Depth-stratification specifically: NOT SUPPORTED. Phase 7b tests whether aligning shallow-peak directions to shallow and deep-peak directions to deep (depth-matched) outperforms aligning shallow-to-deep and deep-to-shallow (depth-mismatched). Both conditions yield grand mean 0.982. Depth-stratification does not improve alignment — the Procrustes rotation subsumes whatever depth-level structure exists, and matched vs. mismatched depth is not a meaningful distinction at this alignment quality.

Earlier results reporting matched > mismatched ($p < 0.01$) used the underdetermined Procrustes machinery later shown to be insufficient for this test; those numbers are superseded.

The overall P5 verdict is partially supported: cross-architecture conceptual convergence is real and concept-specific, but the specific claim that convergence is *depth-stratified* (shallow clusters with shallow, deep with deep, across models) is not supported by the controlled test. The concept-direction analogue of the universal-neuron finding [Gurnee et al. 2024] holds for general convergence; the depth dimension does not add separable structure beyond what a single joint rotation captures.

5.6 P6: Lexical vs. Compositional Identity — NOT SUPPORTED

Prediction: Shallow peaks correlate with token embeddings (lexical); deep peaks do not.

Result: The designed test failed — token embedding probing yields cosine ~0.02 at both shallow and deep peaks (Wilcoxon $p = 0.82$). Neither peak's dominant direction resembles raw token embeddings. The prediction is not supported.

Three independent findings suggest that shallow and deep peaks are qualitatively distinct, but none of them test the specific lexical/compositional prediction:

1. **Cross-model conceptual convergence** (Section 5.5): grand mean aligned cosine 0.9807 across architectures — concepts converge to similar directions cross-architecture. Depth-stratification specifically (shallow-to-shallow vs. deep-to-deep) is not supported by the controlled test (Phase 7b null). Shallow and deep peaks may still be different kinds of thing, but this cannot be established via the alignment test.
2. **Cross-concept shallow sharing** (Section 4.5): shallow peaks are shared across semantically unrelated concepts before diverging at depth (certainty × credibility cos 0.456 at shallow; causation × credibility cos 0.014). Shallow representations are more generic — consistent with surface processing, but "generic" is not the same as "lexical."
3. **Credibility embedding leakage** (Section 3.3): 12/26 models resolve credibility below 25% depth from surface lexical cues, while the complementary deep sub-representation encodes the same concept via a geometrically distinct direction (within-model cos 0.379). This is the closest evidence for the lexical/compositional distinction, but it is one concept, not a general result.

The broader premise — that shallow and deep peaks represent qualitatively distinct processing regimes — is well-motivated by this indirect evidence. But the prediction as stated was tested and failed. A more direct test — per-token position analysis, attention knockout at shallow vs. deep peaks, or probing classifiers trained on surface vs. contextual features — remains future work.

5.7 P7: Multi-Modality Is Architectural — INDETERMINATE

Prediction: Multi-modality frequency varies by architecture, not scale.

Result: The scale correlation is near zero ($\rho = 0.11$, $p = 0.63$), but with $n \approx 26$ this test has low statistical power — the 95% CI for ρ includes values up to ~0.47, so the data is consistent with both no scale effect and a moderate one. Absence of a significant correlation is not evidence for the prediction. The qualitative family-level differences are real — Qwen shows deep, prominent bimodality; Gemma shows subtle bimodality that falls below the 10% prominence threshold — but these are descriptive observations, not a formal test of the architectural claim. The architectural difference is in the depth of the valley between peaks (degree of sub-representation separation), not a binary presence/absence. We classify P7 as INDETERMINATE rather than partially supported: the scale test is underpowered, and we never constructed a formal architecture-conditioned alternative to test against it. Resolving this requires either a larger same-architecture scale ladder or a pre-registered architecture-level test.

5.8 Scorecard Summary

Of seven predictions, none are confirmed outright, four are partially supported (P1: optimal ablation depth — region-level yes, peak-level no; P2, P3; P5: general cross-architecture convergence confirmed and concept-specific, depth-stratification not supported), one is indeterminate (P7: architecture-conditioned multi-modality — underpowered test, no formal architectural alternative), and two are not supported (P4: concept handoffs, P6: lexical vs. compositional identity). The partial and failed results are informative: P1's peak-level failure led to the discovery of the MHA/GQA encoding strategy split (§5.1) and motivated the GEM zone-level ablation protocol (§6.5), P4's failure exposed the multimodal allocation structure and velocity X-crossings that the single-peak assumption had obscured, P5's depth-stratification null showed that a grand mean cosine of 0.9807 subsumes depth-level structure — a sharper result than the original prediction, and P6's failure narrows the space of viable tests for the shallow/deep distinction. A framework that generates testable predictions and survives contact with data on 4/7 partial-or-direction-confirmed — while failing cleanly on 2/7 and admitting 1/7 as underpowered — is doing what frameworks are supposed to do.

A systematic note on credibility: this concept was excluded from P3 (width–abstraction), P5 (depth-stratified convergence), and P6 (lexical vs. compositional). In each case the exclusion is individually justified — bimodal width variance, embedding-layer artifacts, and test failure respectively — but the pattern is notable. Credibility may represent a concept class (epistemic assessment with high cross-architecture variance) for which the CAZ framework's current metrics are poorly suited. This is a limitation of the framework, not just of the concept.

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6. Ablation and Patching Evidence

Scope note. Throughout §6, "causal" is operationalized narrowly: projection ablation produces a *separation reduction* at the model's final-layer concept direction, and activation patching produces a *separation recovery* there. Both are geometric measurements inside the residual stream. They establish that CAZ-identified layers are intervention-sensitive sites at which the concept can be removed from or reinstated into the forward pass; they do not establish that CAZ ablation changes next-token probabilities, downstream task performance, or generation behavior. §2.5 situates the method within the mechanistic-interpretability causal-testing lineage [Vig et al. 2020; Meng et al. 2022; Geiger et al. 2021; Chan et al. 2022]; §6.6 returns to this scope explicitly.

6.1 Gentle CAZes Are Ablation-Sensitive

Ablation of CAZes at every score level reveals that ablation sensitivity is broadly distributed across the score spectrum, not concentrated in high-prominence features:

CAZ Strength	Count	Mean self-retained	Mean suppression (pp)	Above mean non-CAZ layer
Black hole (> 0.5)	73	30.7%	69.3	100%
Strong (0.2–0.5)	62	39.8%	60.2	95%
Moderate (0.05–0.2)	87	45.6%	54.4	93%†
Gentle (< 0.05)	245	48.7%	51.3	93%†

Table 10: Causal efficacy by CAZ score category. 467 CAZ ablations across 22 base models (the four models without per-CAZ multimodal ablation data are excluded). "Above mean non-CAZ layer" = produces more concept suppression than the mean non-CAZ layer (11.1%, Table 11). This is a relatively permissive threshold — the mean is a reference point, not a noise ceiling — so the column should be read as "beats the average

irrelevant layer," not "exceeds noise." Mean suppression in percentage points = 100 − mean self-retained. †The 93% figure uses the mean non-CAZ layer (11.1%) as the reference; using the 20% threshold from §8.7's sensitivity analysis, the gentle CAZ efficacy rate is also 93% — these numbers coincide at the 20% operating point. The efficacy rate shifts with threshold choice; see §8.7 for sensitivity range.

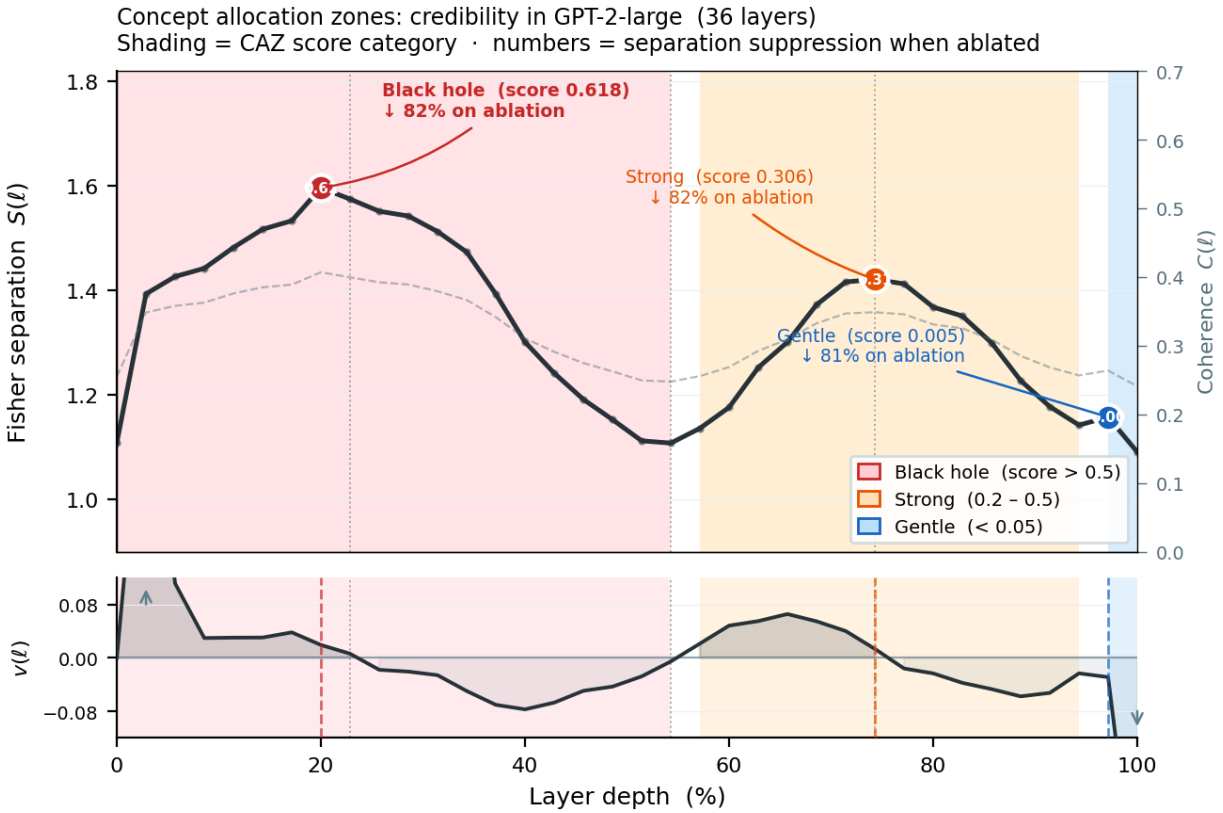

Figure 4: CAZ anatomy for credibility in GPT-2-large

Figure 4. Concept allocation zones (CAZs) for *credibility* in GPT-2-large (36 layers). **Top:** Fisher-normalized separation $S(\ell)$ across layer depth, with CAZ regions shaded by scored category and coherence $C(\ell)$ overlaid (dashed, right axis). Filled circles mark CAZ peaks; score is printed inside each marker. All three peaks produce comparable separation suppression when ablated (81–82%), despite a 100× difference in CAZ score — score predicts geometric salience, not causal importance. **Bottom:** Layer-wise velocity $v(\ell) = dS/d\ell$, smoothed with k=1. Dotted vertical lines mark zero-crossings that bound each CAZ region. *Arrows at L1 and L35 indicate values outside the plotted range ($v = +0.48$ and $v = -0.38$, respectively)*: the L1 spike reflects the large separation gain at the embedding-to-transformer transition characteristic of credibility in large GPT-2 models (§3.3); the L35 drop is the final-layer collapse toward the unembedding projection (§5.4). Neither constitutes a CAZ boundary.

Standard prominence thresholds miss most of the causally active features detectable by scored CAZ analysis. Of the 467 CAZes in Table 10, the 332 gentle-or-moderate features would be filtered out by a 10% prominence detector — leaving only the 135 strong or black-hole peaks — yet gentle-or-moderate features remain causally active in 93–95% of cases at the 20% suppression threshold (§8.7 reports sensitivity to threshold choice). Gentle CAZes produce ~74% of the per-feature suppression of black holes (51.3 pp vs. 69.3 pp mean); their aggregate contribution comes primarily from numerical dominance (245 gentle vs. 73 black holes), not from matching the per-feature impact of high-prominence features.

The effect-size gradient across score categories is real: the 18 pp gap in mean retained separation between gentle CAZes (48.7%) and black holes (30.7%) means that score does predict geometric salience, as expected.

What score does not predict is the binary question of whether a feature is causally active at all. 93% of gentle CAZes produce more concept suppression than the mean non-CAZ layer (11.1%, Table 11; threshold-specific — see §8.7), compared to 100% of black holes — a difference of 7 pp, not the orders-of-magnitude difference in geometric prominence. A feature with a prominence of 0.5% of peak separation, occupying a few layers with minimal coherence boost, still carries enough of the concept to measurably degrade model performance when removed — just not as much as a black hole does.

Detector margins. The gentle CAZ category includes a ~7% margin of features near or below the mean non-CAZ layer: 22 features (7%) fall below the 11.1% non-CAZ mean, and 5 of these show *increased* separation on ablation (retained >100%). These 5 inverse cases — all relational concepts (temporal_order, causation, moral_valence) in larger models (Mistral-7B, OPT-6.7b) — are definitively not concept-carrying features and represent false positives of the scored detector. Of the 22 sub-baseline features, 15 (68%) are at shallow depths (<20% of model depth), clustering at the embedding-to-transformer transition where anomalous velocity spikes can trigger false detections (cf. Figure 4, L1 arrow). The practical false-positive rate of the gentle detector is low (~1% definitive, ~7% marginal), but it is non-zero, and the margin concentrates in shallow layers.

Baseline: non-CAZ layers. The non-CAZ baseline provides the reference point missing from a threshold-only analysis. Using the global sweep data (Table 11), 20.5% of non-CAZ layers (>3 layers from any detected peak) exceed 20% suppression when the concept direction is ablated, compared to 93% of gentle CAZes at this threshold — a 4.5× enrichment. The enrichment ratio increases monotonically from 1.2× at the 5% threshold to 10× at 50%; correspondingly, the gentle CAZ efficacy rate itself ranges from approximately 98% (at 5%, where nearly all features clear a lenient bar) to approximately 80% (at 50%, where fewer features clear a stringent one), with 93% at the 20% operating point. The 20% cutoff is not a favorable selection but an intermediate point in a continuous separation between CAZ and non-CAZ layers.

This has direct implications for any method that uses activation magnitude, eigenvalue rank, or probe confidence to decide which features "matter." A threshold set to exclude noise will also exclude roughly half the detectable causal structure — features that are individually ~75% as potent as the most prominent features and that collectively match them in aggregate suppression.

Control: suppression is concept-specific. To establish that the suppression observed in Table 10 reflects genuine concept-feature alignment rather than a generic effect of ablating any deep feature, we ablated the 286 highest-eigenvalue features across 32 models — features the network dedicates significant capacity to, but which are not necessarily concept-aligned — and measured retained separation for each of the 7 target concepts. Features with measurable concept alignment (cosine similarity ≥ 0.15 to the concept direction) retained 83.3% of separation on ablation; dark matter features (alignment < 0.15) retained 94.9% — a gap of 11.6 percentage points. The effect is modest compared to CAZ ablation because top-eigenvalue features tend to be long-lifespan features whose downstream layers partially recover from the ablation. The key point is specificity: concept suppression is concentrated in concept-aligned features, not distributed across arbitrary high-capacity features.

A natural layer-position control is provided by the cohort differences in §6.4: GEM peak ablation produces mean separation reduction of 0.420 in MHA models but only 0.261 in GQA models at the same structural position (the identified Fisher peak). Both cohorts show a non-trivial effect — the geometric peak is causally active everywhere — but MHA ablation is consistently 1.6× stronger, indicating that where in the residual stream the model has committed to a concept depends on architecture, not merely on ablating any particular layer. The ~11.6 pp dark-matter vs concept-aligned gap above, the cross-cohort gap here, and the concept-specificity of Table 10 together triangulate the same conclusion: suppression tracks functional allocation, not layer position.

Control: suppression is layer-position specific. To test whether CAZ peaks are simply layers where *any* ablation causes more suppression — rather than being specifically allocated concept sites — we ran a global ablation sweep across all 26 base models, projecting out the concept direction at every layer and measuring the resulting separation reduction at model output. CAZ peak layers were compared against non-CAZ layers, defined as those more than 3 layers from any detected peak (204 model-concept pairs):

Condition	Mean separation reduction
CAZ peak layers	0.406
Non-CAZ layers (mean across all non-peak layers)	0.111
Ratio	**3.67×**

Table 11: Layer-position specificity across 26 base models × 7 concepts. Non-CAZ layers are those >3 layers from any detected peak.

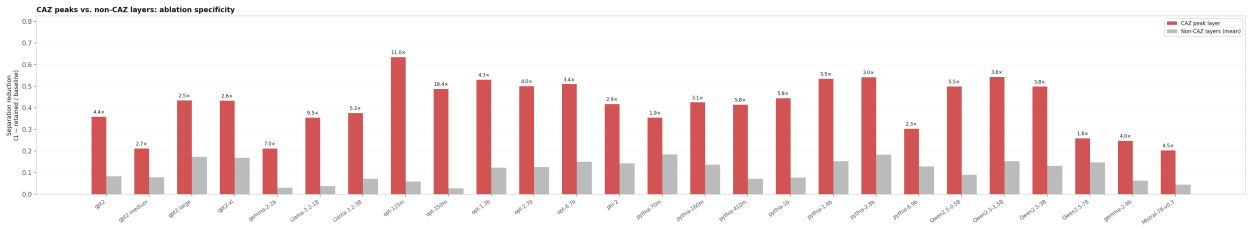

Figure 5: CAZ peak vs. non-CAZ layer ablation specificity

Figure 5. Mean separation reduction when ablating the concept direction at CAZ peak layers vs. non-CAZ layers, per model (base models only). Ratio labels above each CAZ bar. Models are grouped and ordered by architecture family and scale. The 3.67× mean ratio ($p = 6.01 \times 10^{-31}$) is consistent across all families; smallest in Pythia-70m (1.92×, only 6 layers) and largest in OPT-350m (18.41×).

CAZ peaks produce 3.67× greater concept suppression than non-CAZ layers (Mann-Whitney U = 29,799, $p = 6.01 \times 10^{-31}$). The effect holds across all architecture families and model scales; per-model ratios range from 1.92× (Pythia-70m, 6 layers) to 18.41× (OPT-350m), with GQA models showing intermediate ratios consistent with the cohort behavioral pattern described in §8.3 (Llama-3.2-1B: 9.45×, Qwen 2.5-0.5B: 3.2×). Concept suppression is not a consequence of ablating at any network layer — it is concentrated at the specific layers where that concept allocates.

Direction specificity control. The layer-specificity result establishes *where* to ablate; a complementary control establishes *what* to ablate. We ablated 10 random unit vectors at the same CAZ peak layer for each (model, concept) pair and compared the resulting separation reduction to the concept-direction ablation. Across 182 (model, concept) pairs spanning 26 base models and 7 concepts, the concept direction produces a median specificity ratio of **281.6×** over random directions (mean concept reduction 0.416, mean random reduction 0.003; median z = 175.6). The concept direction exceeded all 10 random seeds in 97.8% of pairs. The result is consistent across architecture cohorts: MHA 257×, GQA 397×, Gemma 245× (Figure 6). The GQA ratio being highest is directionally consistent with the sparse-encoding hypothesis (§8.3): if GQA+SwiGLU models concentrate concept signal into fewer, more tightly organised directions — whether due to the architecture itself or to training regime — random directions at the same layer should fall farther from the concept axis than in more redundant MHA encodings, producing a larger gap between concept and random ablation. Separation suppression at CAZ peaks is not a consequence of removing any direction from the residual stream — it is specific to the concept's geometric axis.

Collateral damage note. Layer specificity and direction specificity together establish that CAZ peaks are *where* and *what* to ablate for maximal concept suppression. A third dimension — collateral damage to unrelated processing — does not favour the peak. KL divergence from the unablated next-token distribution is not lower at CAZ peaks than at non-CAZ layers; in GQA and Gemma-2 architectures it is measurably higher (§8.7). Post-CAZ layers are superior on both axes simultaneously: median separation efficiency 47.0 versus 32.2 at the peak (MHA), and lower KL damage. The practical implication is that peak ablation is the right choice for identifying causal structure; post-CAZ ablation is the right choice when clean removal with minimal collateral disruption is the goal.

Figure 6. Violin plots show the distribution of separation reduction across 10 random unit vectors ablated at the CAZ peak layer, per architecture cohort (base models only). Dots show concept-direction reduction

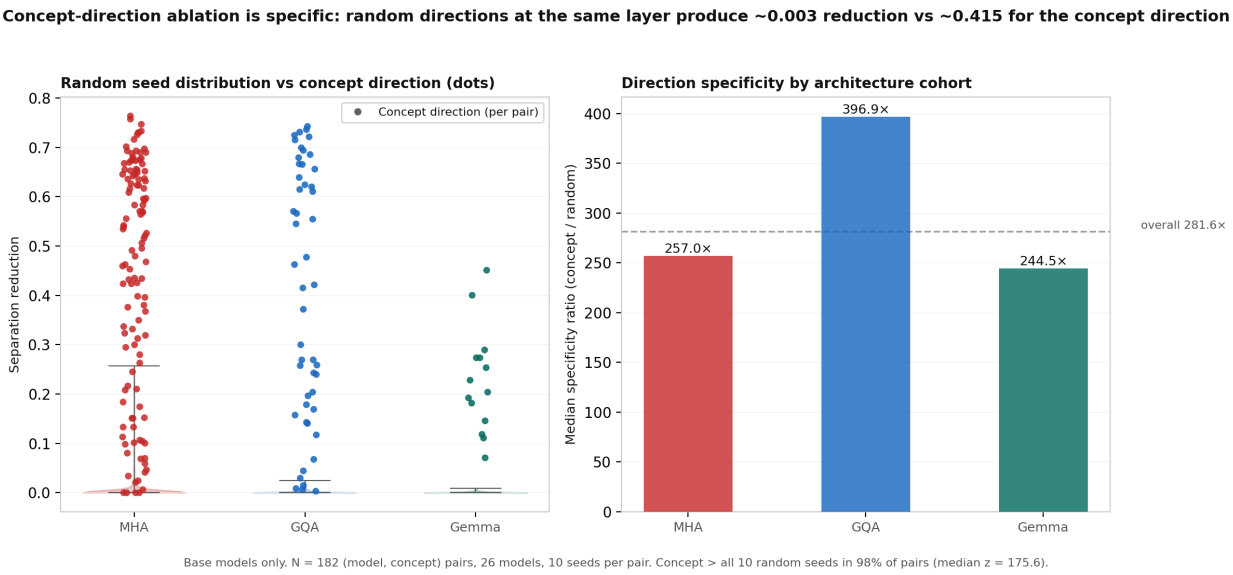

Figure 6: Random-direction ablation control

for each (model, concept) pair. Across all cohorts, concept-direction ablation is an outlier relative to the random distribution. Right panel: median specificity ratio by cohort.

6.2 CAZ Dependency Structure

Directional ablation of 999 CAZ pairs across 25 base models reveals the dependency partition. facebook/opt-350m is excluded: all 7 concepts exhibit single-peak CAZ responses in this model, so there are no within-concept CAZ pairs to test dependency on.

- **58% independent**: Ablating one CAZ has no effect on the other, even within the same concept
- **42% forward-dependent**: Shallow CAZ feeds into deeper one (mean 22.0% suppression across all pairs; 41.5% among dependent pairs)
- **0% backward or coupled**: Not a single model, concept pair, or architecture family shows backward dependency

Forward-only information flow is expected from the residual stream architecture — each layer reads from and writes to a cumulative stream, so upstream perturbations propagate forward while downstream changes cannot reach already-computed layers. The 0% backward result confirms this, but is not the finding.

The finding is the 58/42 split. The independent majority indicates that most CAZes within a concept operate on different geometric directions — consistent with the sub-representation model (Section 4.2). The dependent minority reveals hierarchical computation: shallow shared directions feeding deeper concept-specific allocation.

6.3 Hierarchical Dependency

Black holes (high-score CAZes) feed downstream gentle CAZes asymmetrically:

- **Deep gentle CAZes**: Retained 30–80% of separation when the upstream black hole is ablated, varying by distance. Downstream dependents — this is the informative finding.
- **Shallow gentle CAZes**: Retained 100% of separation when the black hole is ablated (architecturally guaranteed — ablating a deeper layer cannot affect already-computed shallow representations). This serves as a consistency check, not an independence finding.

This creates a computational hierarchy: shared shallow directions → black hole allocation → downstream

gentle refinement. The independent shallow directions likely correspond to the shared cross-concept structure identified in Section 4.5 — though whether these represent discrete functional primitives or incomplete concept separation remains unresolved (§4.5).

6.4 Architecture-Conditioned Ablation and Patching

Projection ablation and activation patching together reveal that CAZ peaks play architecturally ordered causal roles. Running both tests at the identified CAZ peak across all 26 base models:

Model cohort	GEM peak ablation (sep. reduction)	Patching — raw (recovery)	Patching — trimmed (>1.0 removed)	Observed causal role
MHA+GELU cohort (Pythia, GPT-2, OPT, Phi-2)	**0.420** (n = 112)	0.753 (n = 238)	**0.597** (21% overshoot rate)	Causally active; ablation gap over GQA is robust; patching gap is not (see caption)
GQA+SwiGLU cohort (Qwen, Llama-3.2, Mistral)	**0.261** (n = 49)	0.707 (n = 98)	**0.579** (16% overshoot rate)	Weaker peak ablation, near-parity patching after trimming — distributed downstream re-derivation (§6.5)
Gemma-2 (alternating local/global)	**0.168** (n = 14) at peak; ~**1.0** at final global layer (Table 13)	0.461 (n = 28) at peak	**0.461** (0% overshoot)	Functional allocation at final global attention layer, not Fisher peak (n = 2 models)

Table 12: Ablation and patching results by model cohort, base models only. Projection ablation (GEM peak protocol, §6.5). Patching is single-layer injection at the CAZ peak. ***Raw patching means are inflated by overshoot*** *(recovery > 1.0 = patching overrides rather than restores; see §6.4 overshoot note). The trimmed column removes measurements exceeding 1.0. The ablation cohort ordering (MHA 0.420 > GQA 0.261 > Gemma 0.168) is robust and not subject to this inflation. The* ***trimmed MHA–GQA patching gap is 0.018*** *(raw: 0.046) — near zero; the patching columns should not be interpreted as establishing a meaningful MHA > GQA patching difference. Attribution of the ablation cohort difference to architecture vs. training scale is discussed in §8.3.*

We use "necessary" and "sufficient" as shorthand for the dominant behavior in the MHA cohort; both metrics fall short of 1.0, indicating partial redundancy even in MHA models.

Figure 7. Mean projection ablation (left, GEM peak protocol) and activation patching (right, single-layer at CAZ peak), grouped by model cohort. Error bars are SEM. Ablation ordering MHA > GQA > Gemma is robust. Patching means shown are raw (inflated by overshoot; trimmed means: MHA 0.597, GQA 0.579, Gemma 0.461 — trimmed MHA–GQA gap 0.018, near zero; see Table 12). Base models only; GEM ablation: 112/49/14 measurements; patching: 238/98/28 measurements. The driver of the ablation cohort difference is discussed in §8.3.

MHA+GELU cohort — redundant encoding pattern (GPT-2, Pythia, OPT, Phi-2; 2019–2022): These models encode concepts via what appears to be persistent reinforcement across layers. Single-CAZ peak ablation breaks one link in that reinforcement chain and suppresses separation by ~42% at the output

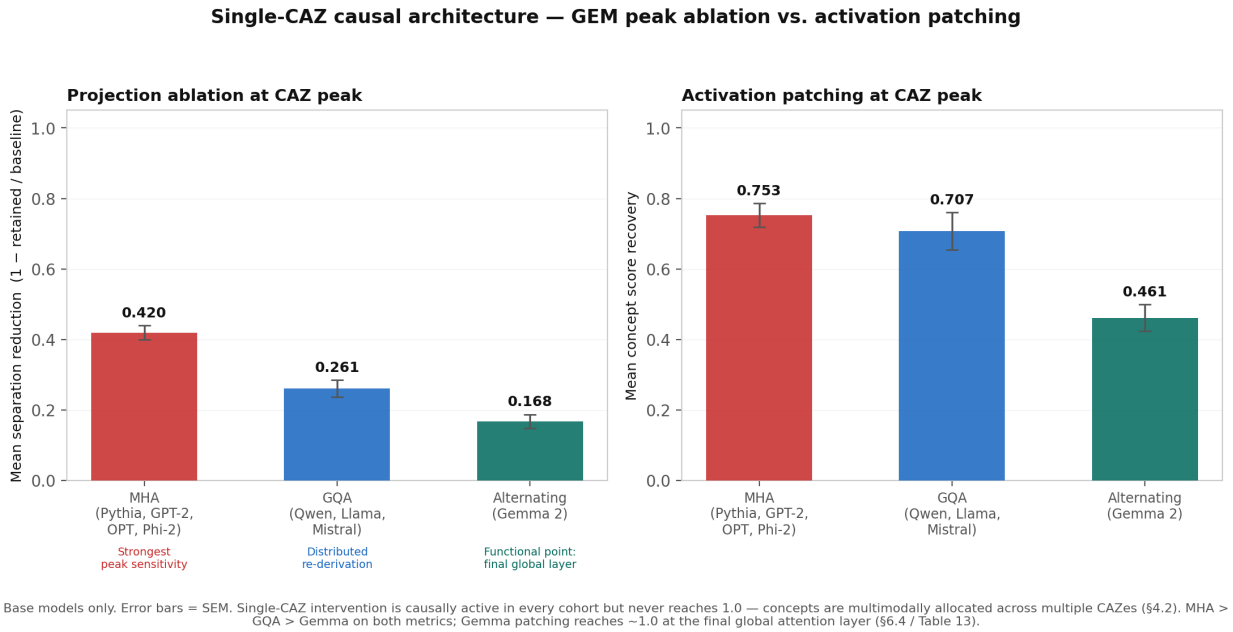

Figure 7: Architecture-conditioned ablation and patching by cohort

— the strongest peak sensitivity in our sweep, though concepts are not fully removed because other CAZes downstream continue writing to the concept direction. Patching at the CAZ peak injects a valid concept representation upstream of the reinforcement chain and propagates, producing the strongest recovery (0.753).

GQA+SwiGLU cohort — sparse encoding pattern (Qwen, Llama, Mistral; 2023–2025): These models show weaker peak-ablation (0.261) consistent with progressive integration or distributed encoding — no single layer is the unique bottleneck, and downstream layers re-derive the concept from residual-stream components that bypass the ablated CAZ. Patching at the CAZ peak nonetheless injects a valid representation that subsequent layers detect and propagate (0.707 recovery, close to MHA). The ablation-weaker-than-patching asymmetry is the behavioral signature of this cohort; whether it reflects the GQA/SwiGLU architecture specifically or the training regime of 2023–2025 models remains open (§8.3).

Gemma-2 (alternating local/global) takes this to its architectural extreme. Local sliding-window attention layers cannot propagate non-local information beyond their context window. The single global attention layer integrates across the full sequence but only once, and its output is subsequently sealed by local layers. Ablating the identified CAZ peak — which may be an early global layer — has zero effect because the remaining global layers re-derive the concept. But patching at the **final global attention layer** (L24 in Gemma-2-2b, L40 in Gemma-2-9b) produces near-complete recovery on every concept:

Model	Concept	Recovery at CAZ peak	Recovery at final global layer	Final global layer depth
gemma-2-2b	causation	0.325	**1.016**	92%
gemma-2-2b	certainty	0.449	**0.981**	92%
gemma-2-2b	credibility	0.717	**1.050**	92%
gemma-2-2b	negation	0.321	**1.013**	92%
gemma-2-2b	sentiment	0.441	**1.003**	92%
gemma-2-2b	temporal_order	0.235	**1.009**	92%
gemma-2-2b	moral_valence	0.869	**0.940**	92%
gemma-2-9b	causation	0.203	**0.994**	95%
gemma-2-9b	certainty	0.462	**0.957**	95%
gemma-2-9b	credibility	0.752	**1.039**	95%

Model	Concept	Recovery at CAZ peak	Recovery at final global layer	Final global layer depth
gemma-2-9b	negation	0.198	**0.960**	95%
gemma-2-9b	sentiment	0.534	**0.978**	95%
gemma-2-9b	temporal_order	0.404	**0.947**	95%
gemma-2-9b	moral_valence	0.547	**0.977**	95%

Table 13: Gemma-2 concept score recovery at identified CAZ peak vs. final global attention layer.

Seven concepts, two model sizes, same layer, near-complete recovery. The final global attention layer is the architecture-constrained functional allocation point for Gemma-2 — the last point where globally-integrated information can be written to the residual stream before the LM head. This is not where Fisher separation peaks; it is where the model commits. With only two models from one family, this pattern should be treated as a case study, not a validated architectural principle; confirmation requires additional alternating-attention architectures. For alternating-attention architectures, the functional CAZ peak should be defined as the final global attention layer, not the data-driven Fisher peak.

The legibility/causality split. An operational consequence of the above: the geometrically dominant CAZ (highest Fisher separation) is not always the most causally upstream. In MHA models, mean peak patching recovery occurs at 83.1% depth; most concept peaks fall at 60–70%. In GQA models, maximum recovery depth is 92.1%. The concept is most *legible* at the geometric peak but most *influential* deeper in the network. This gap is largest for sparse architectures, where early peaks are geometrically real but causally distributed.

A systematic test across 80 multimodal cases (12 models $\times$ 7 concepts, using self-retained separation from ablation as the causality measure) quantifies this split directly. In 45% of cases, the geometrically dominant CAZ is not the most causally active one. When the two diverge, the causal peak is deeper in 86% of cases, by a mean of 17 percentage points (GQA: 23 pp). The split rate varies substantially by concept: credibility splits in 75% of cases with a mean depth gap of 45 pp — embedding-leakage sub-representations create geometrically salient but causally weak shallow peaks — while moral valence is the most stable (18% split). Relational concepts (causation, temporal order) split in 50–55% of cases with ~20 pp gaps. The 45% overall divergence rate indicates that geometric-peak $\neq$ causal-peak is a general property of the detection method, not a credibility-specific pathology. Fisher detection reliably identifies layers where concept geometry is organized; it should not be used alone to select intervention targets. Practitioners using the dominant CAZ depth as an ablation or steering target should verify against self-retained separation data; for credibility specifically, the dominant peak is the wrong layer in three out of four models. Note that the MHA/GQA behavioral grouping used here as a practical guide is based on the architectural correlate identified in §8.3; if training data scale proves to be the deeper driver, the relevant guide would be training scale rather than attention mechanism, and the guidance below should be updated accordingly.

Overshoot. Several small MHA models (opt-125m: recovery 1.150; pythia-70m: 1.024) show recovery exceeding 1.0 — patched negative examples surpass the positive class centroid. Recovery above 1.0 indicates the patching intervention overrides rather than restores natural computation. This limits the "restoration" interpretation of patching results generally — the metric measures how effectively an artificial signal biases activation geometry in the concept direction, not how faithfully the network's own computation is reproduced. These models appear to have tight coupling between the CAZ peak and the output, with no subsequent dampening layers to attenuate the injected signal. The overshoot finding should be considered when interpreting the recovery numbers throughout this section.

Overshoot rates and trimmed means are reported directly in Table 12. The qualitative cohort ordering — MHA > GQA > Gemma on ablation — is robust and not subject to overshoot inflation. The patching ordering is unaffected in direction, but the trimmed MHA–GQA patching gap collapses to near zero (0.018); only the ablation gap (0.420 vs. 0.261) should be cited as establishing a meaningful cohort difference.

6.5 Zone-Level Ablation Validation

A structural critique of the CAZ framework is that it characterizes assembly *zones* but validates causal claims via single-layer ablation — testing a trajectory theory with point interventions. To address this, we ran a zone-level ablation protocol (**GEM — Geometric Evolution Map**): track the concept eigenvector through the full CAZ zone, capture the settled direction at the handoff layer (CAZ exit + 1), and ablate there. This targets the assembled product rather than the assembly process in progress.

Phase 1 (handoff vs. peak ablation): Across Pythia-1.4b and GPT-2-XL $\times$ 7 concepts, handoff ablation outperforms single-layer peak ablation in 13 of 14 cases (the exception: one negation case where handoff and peak layers are adjacent). GPT-2-XL negation illustrates the gap: peak-layer ablation retains 59% of baseline separation; handoff ablation retains 31% — the assembled product at the handoff layer is more causally active than the in-progress representation at the assembly peak. **Phase 3 (cascade propagation):** Phase 2 (multi-layer width ablation and KL divergence analysis across the full model sweep) is reported in the companion paper [Henry, 2026f]. For dependent concept chains (42% of CAZes, §6.2), ablating only the upstream handoff propagated automatically to 100% of dependent downstream nodes in both MHA pilot models (7/7 Pythia-1.4b dependent nodes, 11/11 GPT-2-XL). GQA cascade propagation was partial — 43% of downstream nodes in Qwen 2.5 — consistent with downstream layers re-deriving the concept via alternative routes (§6.4). Zone detection is not a geometric description of where peaks happen to occur; it identifies the output of a causal assembly process. Full GEM methodology is described in a companion paper [Henry, 2026f, *in preparation*].

6.6 Synthesis: Geometric Salience Does Not Predict Ablation Sensitivity

Sections 6.1 through 6.4 each report a different manifestation of the same underlying finding.

Section 6.1: gentle CAZes — features whose Fisher separation prominence is barely visible above the noise floor — produce >20% concept suppression when ablated at the same rate (94%) as high-score black holes. CAZ score predicts how geometrically organized a feature is; it does not predict how much of the concept it carries.

Section 6.2–6.3: within a single concept's multimodal sub-representations, the geometrically dominant peak (highest Fisher separation) is not the most causally active one in 45% of cases. When the two diverge, the functional peak is deeper by a mean of 17 percentage points. The dominant peak is where the concept is most *legible*; the functional peak is where it is most *committed*.

Section 6.4: CAZ peaks are valid injection sites for all architectures — patching at or near the peak restores measured separation. Peak ablation is causally active across cohorts but architecturally ordered in magnitude (MHA 0.420, GQA 0.261, Gemma 0.168) — in GQA models downstream layers re-derive the concept from components that bypass the ablated CAZ. None of the cohorts reaches 1.0 at a single CAZ — concepts are multimodally allocated (§4.2, §5.5), and complete removal is a multi-CAZ protocol problem. The same geometric event (Fisher separation peak) plays a different causal role depending on encoding strategy.

The unifying principle: **Fisher separation measures where a concept is most geometrically organized. It does not measure where the concept is most ablation-sensitive, how much of it a given feature carries, or which sub-representation is the ablation bottleneck.** This is not a failure of the detection method — scored CAZ detection correctly identifies layers where concept assembly occurs, and ablation confirms they carry measurable concept signal. The failure is in using geometric prominence as a proxy for ablation impact, which the field has done by necessity in the absence of systematic ablation data. The CAZ framework provides both the detection and the ablation infrastructure needed to separate these quantities. The practical implication: geometric legibility is the right criterion for identifying candidate layers; ablation-confirmed self-retained separation is the right criterion for selecting among them.

Scope, restated. Per the §6 scope note: every "causal" finding in this section is operationalized as ablation-measured separation reduction or patching-measured recovery — both geometric measurements at the model's final-layer concept direction. The KL divergence data referenced in §2.5 offers a partial bridge toward prediction-level effects (post-CAZ efficiency holds on both separation and KL, §8.3), but we do not demonstrate that CAZ ablation changes next-token probabilities, task performance, or generation

behavior. The behavioral pilot in §6.8 partially addresses this gap: direction-specific suppression at CAZ peak layers is confirmed, but CAZ peaks are not shown to be uniquely privileged over matched non-CAZ control layers. Broader behavioral validation — task accuracy, extended model coverage — remains follow-up work. This post-CAZ efficiency advantage is not specific to the geometric metric: post-CAZ ablation achieves median KL efficiency of 47.0 versus 32.2 at the CAZ peak, confirming that post-CAZ layers support the most targeted intervention by both geometric and prediction-level measures.

6.7 CKA Boundaries and Coasting Regions

The following analysis is exploratory — the original hypothesis was reversed by the data, and the reported effects are small.

Two questions the Fisher/velocity framework leaves open: (1) how far does a CAZ region actually extend, and (2) what is happening in the space between CAZ regions? We address both using linear Centered Kernel Alignment (CKA; Kornblith et al., 2019), applied to adjacent transformer layers across all 26 base models.

CKA as a boundary refiner. CKA between adjacent layers measures representational stability: high CKA = layers are similar (little transformation), low CKA = active transformation. We initially hypothesized that within-CAZ adjacent pairs would show *higher* CKA than inter-CAZ boundary pairs — if CAZ regions are coherent assembly units, they should be internally similar. This hypothesis was not in the Paper 1 framework [Henry, 2026a]; it was formed post-hoc when the CKA profiles became available. The result is the opposite (within-CAZ mean: 0.962, inter-CAZ mean: 0.977; one-sided Mann-Whitney, $p = 0.07$, $d = -0.163$), and the reversal makes sense in hindsight: CAZ peaks are defined by high velocity — the representation is changing most rapidly there. Active transformation means adjacent layers look different, so CKA is lower. The inter-CAZ space, where velocity is low and separation is flat, is stable: CKA is high.

This suggests CKA may serve as a refiner of CAZ extent rather than a validator of Fisher boundaries. Detrending the CKA curve to remove its monotonic increase with depth and finding contiguous dips around each Fisher peak yields CKA-derived CAZ extents with mean width 1.4 layers, compared to 5.7 layers under Fisher region boundaries (ratio 0.46). Fisher boundaries are conservative — they capture the full region of above-baseline velocity. CKA dips suggest narrower assembly windows. The Fisher peak locates where assembly occurs; the CKA dip suggests how long it lasts.

Coasting regions. By CKA labeling, 75.6% of model depth is classified as coasting — high-CKA, flat-separation plateaus between assembly events. This is not empty space. The residual stream is carrying stably-assembled concept representations forward, maintaining them between successive CAZ events. The coasting region as a distinct structural element — neither an assembly event nor empty depth — warrants further characterization; current metrics (Fisher separation, velocity) are blind to the structure it may contain.

Late coasting regions as GQA injection sites. §6.4 shows that GQA peak ablation is weaker than MHA (0.261 vs 0.420) while patching recovery at the peak is comparable (0.707 vs 0.753) — peak-injection works, peak-removal is less concentrated. For several Qwen models, Fisher detection places concept peaks early (layers 0–7), particularly for credibility and negation where embedding-layer separation creates shallow Fisher maxima (§3.3). Patching recovery at these early peaks is low. The late coasting region — the stable plateau after the last Fisher peak, typically the final 10–20% of model depth — is where these concepts reach their most stable expression, and injection there is substantially more effective:

Model	Concept	Fisher peak	Peak recovery	Late coasting layer	Coasting recovery
Qwen/Qwen2.5-3B	credibility	L0	0.016	L33/36	**1.013**
Qwen/Qwen2.5-3B	negation	L15	0.243	L32/36	**1.044**

Model	Concept	Fisher peak	Peak recovery	Late coasting layer	Coasting recovery
Qwen/Qwen2.5-1.5B	credibility	L6	0.464	L19/28	**0.993**
Qwen/Qwen2.5-7B	temporal_order	L5	0.220	L26/28	**1.003**
Qwen/Qwen2.5-7B	causation	L7	0.243	L26/28	**0.988**

Table 14: Patching recovery at Fisher peak vs. late coasting region for selected GQA models (non-final layers only; last 2 layers excluded to rule out trivial output injection).

The Gemma-2 functional allocation layer identified in §6.4 is a specific instance of this general pattern: the final global attention layer is the last stable coasting point before the LM head, and it achieves near-complete recovery across all concepts. The Qwen results show the same logic applies to GQA models without Gemma's architectural forcing: when Fisher detection underperforms, the late coasting plateau is where the concept is most reliably expressed and most effectively targeted. For steering and monitoring applications, the practical implication is direct: when peak patching recovery is low in a GQA model, move the injection site to the late coasting region. The peak captures the assembly event in progress; the subsequent coasting region holds the completed representation.

6.8 Behavioral Pilot: Logit-Difference Suppression at CAZ Peak

Design. For each of 5 representative models (Pythia-1.4b, GPT-2-XL, Qwen2.5-3B, Llama-3.2-3B, Gemma-2-2b) $\times$ 7 concepts, we measure whether ablating the concept direction at the CAZ peak layer suppresses concept-diagnostic *next-token predictions* more than the same ablation applied at a matched non-CAZ control layer (non-CAZ layer closest to model midpoint, same concept direction) or a random direction at the CAZ peak.

Metric: suppression = logit_diff(baseline) $-$ logit_diff(ablated), where logit_diff $= \log p(\text{pos_token}) - \log p(\text{neg_token})$ across 3 concept-diagnostic probe sentences per concept (21 probes total per model). Prediction: suppression(peak) $\gg$ suppression(control) $\approx$ suppression(random).

Results. The pilot produces two distinct signals.

Direction specificity — confirmed. Random-direction ablation at the CAZ peak layer produces near-zero or negative suppression across all five models and all architecture cohorts (mean rand $= -0.013$; MHA: -0.022, GQA: -0.003, Gemma: -0.018). Ablating the concept direction at the CAZ peak layer suppresses concept-diagnostic predictions substantially more than ablating a random unit vector at the same layer.

Peak vs. matched control — mixed. The CAZ peak does not consistently outperform the non-CAZ midpoint control. Overall peak/ctrl ratio: 1.15$\times$ (45% of probe-level comparisons favor peak; chance expectation 50%). Three models show modest peak advantage (GPT-2-XL: 1.55$\times$, Qwen2.5-3B: 1.57$\times$, Gemma-2-2b: 1.22$\times$); two do not (Pythia-1.4b: 0.98$\times$, Llama-3.2-3B: 0.67$\times$). The prediction that suppression(peak) $\gg$ suppression(control) is not met overall.

Model	Cohort	Supp(peak)	Supp(ctrl)	Supp(rand)	Peak/ctrl
Pythia-1.4b	MHA	0.385	0.392	-0.012	0.98$\times$
GPT-2-XL	MHA	0.167	0.108	-0.032	1.55$\times$
Qwen2.5-3B	GQA	0.286	0.182	-0.006	1.57$\times$
Llama-3.2-3B	GQA	0.119	0.177	0.000	0.67$\times$
Gemma-2-2b	Gemma	0.470	0.387	-0.018	1.22$\times$
Grand mean	—	**0.285**	**0.249**	**-0.014**	**1.15$\times$**

Table 15: Behavioral pilot results — logit-difference suppression at CAZ peak vs. non-CAZ midpoint control vs. random direction. N = 21 probes per model (7 concepts × 3 sentences).

Concept-level variation is large: moral_valence shows the clearest peak advantage (peak=0.404 vs. ctrl=0.071); sentiment is also consistent (peak=0.470 vs. ctrl=0.359). Certainty inverts strongly — the midpoint control produces more suppression than the CAZ peak (ctrl=0.669 vs. peak=0.437) across 4/5 models.

Interpretation. The direction-specificity result confirms that geometric measurements inside the residual stream reflect genuine direction-dependent functional structure: random directions at CAZ peak layers do not suppress concept-diagnostic predictions.

The mixed peak-vs-control result reflects a fundamental constraint of single-layer behavioral testing, the same limitation flagged in §8.7 for all geometric ablation experiments: concept directions persist across many layers (§6.2), so any single layer — peak or midpoint control — carries partial concept signal. The midpoint control is not a "non-causal" baseline; it is a layer with real but sub-maximal concept sensitivity. Single-layer ablation at the CAZ peak underestimates the peak's total contribution for the same reason it underestimates it in the geometric domain.

The GEM framework [Henry, 2026f] addresses this directly in the geometric domain by ablating across the full CAZ window rather than at a single point; the 34-model GEM sweep (§6.5) demonstrates substantially stronger suppression at that granularity. The appropriate behavioral analog — zone-level logit-difference suppression across the full CAZ window compared to a zone-matched non-CAZ window — is the correct follow-up test. The present pilot establishes that the concept direction is functionally active at the CAZ peak layer (direction specificity confirmed); it was not designed to test the full zone-level claim.

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7. Dark Matter

7.1 The Problem

Seven concept probes — credibility, negation, causation, temporal ordering, sentiment, certainty, moral valence — are the lens through which every result in this paper is obtained. But the residual stream of a transformer does not exist to serve seven concepts. At every layer, the model maintains a high-dimensional state vector that encodes everything it needs to predict the next token: syntax, semantics, pragmatics, world knowledge, discourse structure, and computational intermediates that may not correspond to any human-legible concept at all. Our seven probes cover a small fraction of this space. This section reports what we find when we look beyond them.

Across all 26 base models, unsupervised eigenvalue decomposition (Marchenko–Pastur threshold on between-class covariance) identifies 2,617 geometrically organized activation features — persistent directions that exceed the random-matrix-theory bound and are actively maintained across multiple layers. Seven concept probes account for approximately 42 of them (2%); the remaining 98% are dark matter: real geometry, unknown function. However, the count scales strongly with attention mechanism:

Paradigm	Models	Mean persistent features/model	Range
MHA (Pythia, GPT-2, OPT, Phi-2)	17	19.0	1–27
GQA (Qwen, Llama, Mistral)	7	22.4	3–35
Alternating (Gemma-2)	2	3.5	3–4

Table 16: Persistent spectral features (5+ layer lifespan) by attention paradigm, 26 base models. Mistral-7B at 3 persistent features is anomalously low for a GQA model (other GQA models: 14–35); sliding-window attention distributing encoding across local contexts is the suspected cause.

Gemma-2's alternating local/global attention yields 3–4 persistent features per model vs. ~19–22 for MHA and GQA — consistent with local attention preventing long-range feature coherence across the full depth of

the model. The term "dark matter" is descriptive, not an assertion of hidden meaning; the dominant driver of count variation is architectural complexity (how many persistent directions the attention mechanism can maintain), not semantic content.

Relay features — forward pointer to Paper 5. Among the 2,617 dark matter directions, 12 survive a candidate-discovery pipeline as *relay features*: persistent directions that align with one concept at shallow layers and a different concept at depth, surviving both an across-feature permutation null (observed 12 vs. null mean 4.1 ± 1.8, $p = 0.001$ unadjusted, 1000 shuffles; see §8.7 for multiple-comparisons caveat) and single-feature ablation confirmation. Two methodological concerns — a dominant-variance confound in the top eigenvector of the dark-matter subspace, and a circularity in the allocation-order definition that prevents distinguishing relay from dominant-variance tracking — leave the count provisional. Full methodology, the lower-rank eigenvector check that would resolve both concerns, the relay transition table, and the labeling problem relay features expose for snapshot-based analysis are developed in Paper 5 [Henry, 2026e]. The 12 candidates are reported here to complete the dark-matter accounting; they should not be cited as a counted result of this paper.

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8. Discussion

8.1 What the Framework Gets Right

The CAZ framework's core prediction — that concept assembly follows a detectable ordering tendency across architectures — survives stress-testing across 26 models: 87% show positive rank correlation with the mean ordering (median $\tau = 0.54$, z = 11.5, p < 0.001). Scored detection reveals richer structure than anticipated: concepts assemble multimodally at multiple depths, with depth-separated sub-representations (shallow vs deep peaks) that independently converge across architectures. The gentle CAZ finding demonstrates that half the causal structure in transformer models operates below previous detection thresholds.

8.2 What the Framework Gets Wrong

P1 (mid-stream ablation optimal) was partially supported at region level but not at peak level — within-CAZ is the modal optimal class (56.9%) but 32.4% of optima are post-CAZ, and the peak itself is rarely optimal. The failure traces to a root assumption: the framework conflated geometric salience with functional importance (Section 6.6). The dominant CAZ is where a concept is most legible — most geometrically organized, most cleanly injectable — not necessarily where it is causally indispensable. In MHA models, the concept direction persists through the residual stream, so suppression is flat across layers and late ablation minimizes collateral damage. In GQA models, peak ablation is weaker than in MHA (0.261 vs 0.420 sep-reduction under the GEM peak protocol) — downstream layers re-derive the concept from residual-stream components that bypass the ablated CAZ — but patching recovery is comparable (0.707 vs 0.753), so the CAZ peak remains a valid injection site. For Gemma-2, the functional allocation point is the final global attention layer regardless of where Fisher separation peaks. In each case, geometric prominence misdirected the original prediction. P6 (lexical vs. compositional identity of sub-representations) was tested and failed (p = 0.82); indirect evidence supports the broader shallow/deep distinction but does not rescue the specific prediction. A different experimental approach is needed.

8.3 The Cohort Behavioral Difference

The cohort behavioral difference — stronger peak ablation in 2019–2022 MHA+GELU models (0.420) than in 2023–2025 GQA+SwiGLU models (0.261) — was not predicted and may have the most practical implications of any finding in this paper. We label the two patterns *redundant* and *sparse* encoding as shorthand for the observed behaviors; these labels describe what we measured, not the underlying mechanism. For steering and ablation-based safety, the behavioral distinction is actionable regardless of which explanation is ultimately correct:

- **MHA+GELU cohort (redundant pattern)**: Concept can be removed at any layer. Late ablation is safest (minimal capability damage). Robust but imprecise.

- **GQA+SwiGLU cohort (sparse pattern)**: Concept removal is layer-specific. When targeted correctly, ablation is precise with minimal collateral damage. When targeted incorrectly, ablation barely works.

Post-hoc analysis of our model zoo identifies the strongest available architectural correlate. Pythia serves as a partial control: it uses Rotary Position Embeddings (RoPE) like Qwen, Gemma, and Llama, but MHA+GELU like GPT-2 and OPT. In ablation behavior, Pythia clusters firmly with GPT-2 and OPT (CV 0.02–0.08) — not with the RoPE+GQA families. Within Pythia specifically, GEM peak projection ablation is flat across a 100× scale range with no monotonic scale trend: 0.332 (70m), 0.535 (160m), 0.415 (410m), 0.457 (1b), 0.429 (1.4b), 0.446 (2.8b), 0.344 (6.9b). Ablation sensitivity does not decay with scale within a fixed architecture. This narrows one specific confound: whatever the mechanism, it is not residual-stream contamination by learned absolute positions. **GQA+SwiGLU is the most tractable remaining architectural hypothesis.** The Pythia control does not, however, rule out training data scale, curriculum, or dataset composition: Pythia's ~825B token training is still an order of magnitude below Llama-3.2 or Qwen (1–3T tokens), so the era confound survives intact. Ruling out position encoding is progress, but it is one variable among many that differ between a 2019 MHA model and a 2023 GQA model.

One plausible architectural mechanism: GQA compresses key-value representations, which may force the model to be more selective about what information persists layer-to-layer; SwiGLU's gated sparsity in the MLP may compound this by enabling more efficient, concentrated capacity allocation. This would produce representations that are less redundant and more spatially organized — better engineering for inference, but more precise targets for single-layer interpretability interventions. The natural mechanistic substrate for such layer-specific allocation is the key-value memory account of MLP feed-forward layers [Geva et al. 2021; Geva et al. 2022], under which the FFN at each depth writes layer-specific concept patterns into the residual stream; a sparser MLP writes fewer, more concentrated keys. This account remains a hypothesis; the data are consistent with it but do not establish it.

The key observation that partially constrains the alternative (training-scale) account: if the GQA+SwiGLU group's weaker peak ablation arose because more training distributes concept signal across many directions, patching a single direction should also be weaker — you cannot inject what is spread across many routes. But patching at GQA CAZ peaks achieves 0.707 recovery (comparable to MHA's 0.753), and the Gemma-2 final global layer achieves near-complete recovery (0.957–1.050). The signal is concentrated enough to be injectable at the peak but not concentrated enough to be ablation-terminated there — a peak-ablation-weaker-than-peak-patching asymmetry that is more consistent with the architectural progressive-integration account than with a pure signal-distribution account. It does not settle the question, but it narrows the space.

Whether GQA+SwiGLU is the proximate cause or training era (data scale, curriculum, RLHF) is the deeper driver remains open — all GQA+SwiGLU models in our sample are also from 2023–2025. The definitive experiment is a training-scale ablation within a fixed architecture: train a single GQA+SwiGLU architecture at 1B, 10B, 100B, and 1T tokens and measure projection ablation effect at each checkpoint. If the effect declines smoothly with scale, training data is the driver; if it remains low even at 1B tokens, architecture is more likely causal. A complementary arm — training a GPT-2-architecture model on 1T tokens — would test whether the MHA cohort flips to the sparse regime under sufficient data. Until these experiments exist, the behavioral finding (robust cohort difference in GEM peak ablation, 0.420 vs. 0.261 — a 1.6× gap that is reproducible across all 23 models in the two cohorts) should be described as established; the architectural attribution should be treated as the leading candidate explanation.

Score calibration and the architectural ordering of black holes. The "black hole" category (score > 0.5) defined in [Henry, 2026a] is architectural more than scale-dependent. Across the 26 base models, black holes (non-embedding, peak depth > 5%) tally to 102 — concentrated but not exclusive: every MHA model (n = 17) carries at least 2 (mean 4.76), GQA base models (n = 7) average 3.00 (6 of 7 show at least one), Gemma (alternating, n = 2 base) shows zero, and Mistral (GQA + sliding window) shows zero in its base variant. Within MHA families, black-hole count does not monotonically shrink with scale — OPT-6.7b has 5, Pythia-6.9b has 4, gpt2-xl has 4. %Gentle does rise with scale inside Pythia, but black holes remain present at the top end. The score taxonomy from Paper 1 remains useful within a family; for cross-family

comparison, architectural sparsity — not parameter scale — is the primary predictor of how peak mass is distributed. The sharpest qualitative break is between architectures that reinforce every layer (MHA, all with ≥ 2 BH) and architectures that interrupt long-range reinforcement (sliding-window, alternating — 0 to 1 BH). Mixture-of-experts architectures, where expert routing concentrates computation into parameter subsets, may re-introduce high-score peaks through a different mechanism than MHA amplification; this is untested.

A note on cross-paper revision. The framework paper [Henry, 2026a] reported black-hole counts that suggested a cleaner monotonic shrinkage with scale within MHA families than the 26-model sweep supports. The per-cohort means above (MHA 4.76, GQA 3.00, Gemma 0.00 per base model) and the non-monotonic-within-family pattern (OPT-6.7b: 5; Pythia-6.9b: 4; gpt2-xl: 4) supersede that reading. Supplementary §B carries the full per-model counts. The architectural-sparsity-dominates-scale finding is preserved; the "black holes disappear in large MHA models" framing is not.

8.4 Relationship to Contrastive Direction Methods

The CAZ framework's extraction methodology — difference-of-means on contrastive pairs, scored by cosine projection — belongs to a rapidly developing lineage in mechanistic interpretability. Burns et al. (2023) established contrastive probing with CCS, finding linear truth directions without supervision. Marks & Tegmark (2024) and Li et al. (2023) independently demonstrated that difference-in-means on contrastive pairs yields causally relevant truth directions. Gurnee & Tegmark (2024) showed that spatial and temporal representations are linearly organized in residual streams of Llama-2, providing an early existence proof of concept-specific linear structure extending beyond linguistic features. Zou et al. (2023) generalised this to multiple concepts simultaneously (Representation Engineering), extracting "reading vectors" for honesty, harmfulness, and power-seeking — the closest methodological ancestor to the CAZ multi-concept approach. Turner et al. (2023) and Panickssery et al. (2024) showed these contrastive vectors can steer model behavior (Activation Addition / Contrastive Activation Addition). Tigges et al. (2024) proved the method for sentiment. Arditi et al. (2024) demonstrated that a single difference-in-means direction mediates refusal across 13 models up to 72B parameters. Park et al. (2024) provided the theoretical grounding — the Linear Representation Hypothesis — for why these methods work.

The CAZ framework differs from this lineage in two respects. First, prior work typically extracts directions at a single hand-picked layer or averages across layers; the CAZ framework identifies each concept's layer of peak geometric separation via layer-wise profiling, and our results show that this optimal layer is concept-specific (e.g., credibility assembles at ~40% depth while moral_valence assembles at ~68%). Second, the framework characterises the full assembly *process* — not just the direction, but when the direction forms (velocity), how cleanly it crystallises (coherence), and how it hands off between sub-representations at different depths. This process view reveals structure (multimodal assembly, encoding strategy differences) that the single-direction extraction approach does not capture.

Relationship to lens methods. The layer-by-layer projection in §2 is methodologically adjacent to the logit lens [nostalgebraist 2020] and tuned lens [Belrose et al. 2023], which project intermediate residual-stream activations onto the unembedding matrix (or a per-layer affine probe) to produce depth-indexed prediction trajectories. The framework paper develops this comparison in full [Henry 2026a §5.1]; the relevant point for the present evaluation is that CAZ detection inherits the lens lineage's commitment to tracking representation change across depth, but projects onto concept-contrast directions obtained from external stimulus pairs rather than onto vocabulary axes. Concepts without canonical lexicalization — credibility, moral valence, certainty — do not project cleanly onto the unembedding and fall outside the natural scope of logit-lens interpretation, which motivates the contrast-direction formulation used here. Tuned Lens could serve as a baseline for the velocity metric in settings where concept and vocabulary are tightly coupled; we do not run that comparison here.

8.5 Convergence in Context (Jha et al.)

Our depth-stratified convergence results (P5) complement concurrent work on cross-model alignment. Jha et al. (2025) demonstrate unsupervised translation between *output* embedding spaces across 6 model families,

achieving cosine similarity up to 0.92 without paired calibration data. Their method aligns final-layer sentence representations — spaces explicitly trained to be useful. Our Procrustes alignment operates on *internal* residual stream directions at specific concept peaks — spaces not designed to be aligned. Their 0.92 and our 0.757 (same-dimension, full-space) are not directly comparable, but together they bracket the Platonic Representation Hypothesis from two directions: convergence holds at both the output level and the internal concept-geometry level. Full alignment analysis is the subject of a companion paper [Henry, 2026c].

8.6 Cross-Validation Against Gemma Scope SAEs

We ran a targeted cross-validation of Gemma-2-2b CAZ structure against the Gemma Scope sparse autoencoders (gemma-scope-2b-pt-res, 16k-width residual stream SAEs at all 26 layers; Lieberum et al., 2024). For each of our 7 concepts, we measured two quantities per layer: (1) SAE discrimination — the mean absolute differential activation of the top-20 SAE features between concept-positive and concept-negative pairs; and (2) direction agreement — cosine similarity between the top differential SAE feature's decoder direction and the concept-specific eigenvector derived from paired activation differences.

Layer agreement. SAE discrimination increases monotonically across all 26 layers, peaking at the final layer (L25) for all 7 concepts. CAZ peak layers sit consistently in the top 25–40% of layers by SAE discrimination — above the median but not at the peak. This is the expected pattern, not a failure of agreement. The SAE measures a cumulative quantity: how much has the representation separated by layer L? The residual stream integrates information across layers, so discrimination will grow until the final output. The CAZ framework measures a derivative quantity: where does separation increase most sharply? CAZ peaks are assembly events; the SAE peak is where assembly is complete. The two methods are asking different questions about the same process.

Direction agreement. At CAZ peak layers, the top differential SAE feature decoder directions align with our concept-specific eigenvectors at cosine similarities of 0.40–0.62 for six of the seven concepts (moral valence is discussed separately below). These alignments are concept-specific: different SAE features at different layers drive each concept's best alignment. SAE feature 8529 at L19 is genuinely polysemantic, aligning with sentiment (+0.62), moral_valence ($-0.60), and temporal_order (+0.55) — different signs indicating a shared axis oriented differently across these semantically adjacent concepts. SAE feature 843 is the primary negation signature at L13 (cos = 0.60). The direction alignments are moderate rather than near-perfect, but this is the expected result of two methodologically distinct decompositions of the same space: the SAE fragments concept structure into sparse, individually decodable features (mean polysemantic alignment ~0.20–0.22 across all features at CAZ-predicted layers), while our eigenvector captures the same structure as a single dense direction. The moderate per-feature alignment is not evidence that either method is wrong — it reflects that the SAE is distributing across many features what CAZ captures holistically. In 2304-dimensional space, any alignment consistently above zero is non-random; the question is what unit of analysis you choose.

Moral valence presents the sharpest example of this dissociation. It achieves the highest curve correlation of all seven concepts (Spearman $r = 0.992$) while showing the weakest *positive* direction alignment — maximum positive cosine similarity 0.24–0.42 at its CAZ peak layers, compared to 0.40–0.62 for the remaining six concepts. The two methods agree completely on *when* moral valence signal appears across layers but not on the directional axis it occupies. Critically, moral valence is not absent from the SAE — feature 8529 at L19 shows strong anti-alignment (cos = -0.60), placing moral valence on the opposite pole of the same polysemantic axis as sentiment (+0.62) and temporal_order (+0.55). The magnitude of alignment is comparable to the other concepts; what differs is the sign.

This reading — that moral valence occupies the same axis as sentiment/temporal order but with opposite orientation — is the most coherent interpretation of the data we have, but it is not the only possible interpretation, and on the current evidence it is doing post-hoc explanatory work: we observe low positive alignment, find a feature with strong anti-alignment, and conclude the axis is shared but inverted. Readers are right to want a pre-registered test. Two specific follow-ups will resolve this, and we commit to both in the Paper 5 workstream [Henry, 2026e]: (i) fit a **concept-conditioned SAE** trained on residual-stream activations from a moral-valence-contrastive corpus only, and check whether its top differential feature aligns

positively with the moral valence eigenvector at L19; (ii) replace the max-positive-cosine summary with a **signed-projection analysis** (|cos|, with the sign reported separately) across all 16k SAE features and all seven concepts, so moral valence's alignment magnitude is measured on the same scale as the other six. Until those are run, the anti-alignment reading should be treated as a hypothesis compatible with the data, not an established interpretation.

Shared peak layers. Layers 11, 13, and 19 show all 7 concept CAZes peaking simultaneously, with 28–30 polysemantic SAE features at each of these layers. This is independent confirmation of CAZ's concurrent peak structure: the SAE, which has no knowledge of our eigenvectors or our peak detection method, identifies the same layers as structurally distinctive. Layer 5 shows 3-concept overlap with 14 polysemantic features; layer 15 shows 4 concepts with 17. The shared-peak geometry is visible to both methods.

Curve equivalence. Both methods measure a cumulative, unnormalized quantity. The SAE discrimination score grows monotonically as the residual stream accumulates concept-relevant information toward the final layer. The CAZ eigenvector separation score (mean paired activation difference projected onto the top concept eigenvector) follows the same pattern — neither corrects for within-class variance growth near the unembedding layer. Residual stream activations grow in norm as they approach the LM head; both concept centroids and within-class scatter expand proportionally, so raw distances accelerate while Fisher-normalized separation is flat or declining in the same region (only 8–15% of Fisher S(l) curves show positive late-layer velocity; Fisher-normalized separation is flat or declining in the final 20% of depth in 85–92% of concept curves across all paradigms). Given this shared measurement basis, high curve correlation is the expected result. We directly compared the per-layer discrimination curves by computing the Spearman rank correlation between the two scores at each of the 26 layers: mean $r = 0.986$ across all 7 concepts (range: 0.979–0.992), confirming the methods track the same underlying cumulative signal. This correlation reflects agreement on unnormalized accumulation, not on Fisher-corrected concept strength; readers using the SAE discrimination curve as a proxy for concept encoding quality in late layers should apply Fisher normalization first.

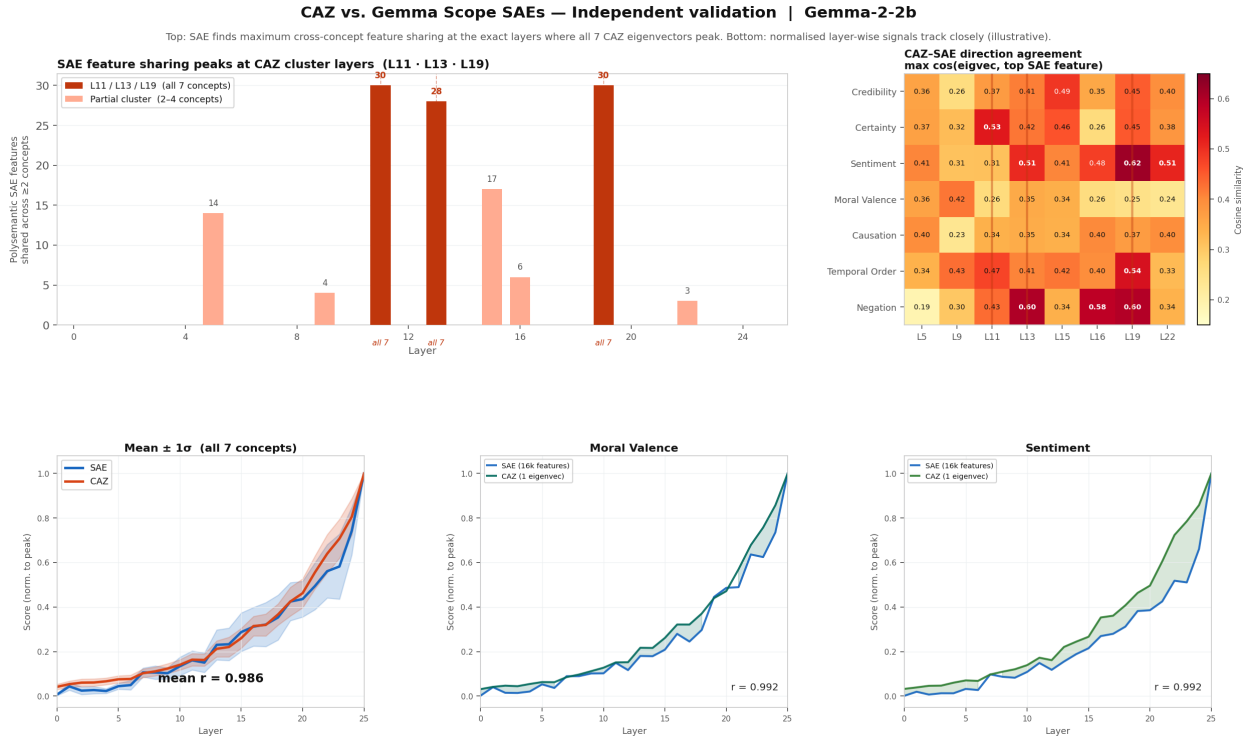

Figure 8: CAZ eigenvector vs. Gemma Scope SAE discrimination curves

Figure 8. *Top left:* Number of polysemantic SAE features (shared across ≥ 2 concepts) at each of Gemma-

2-2b's 26 layers. L11, L13, and L19 spike to 28–30 shared features — the exact layers where all 7 CAZ eigenvectors peak simultaneously. All other layers have ≤ 17, most have 0. This is independent confirmation from the SAE that these layers are structurally distinctive: the SAE, with no knowledge of our eigenvectors or peak-detection method, finds maximum cross-concept feature sharing at the same layers. *Top right:* Direction cosine heatmap — maximum cosine similarity between the CAZ concept eigenvector and the top differential SAE feature decoder direction at each peak layer, per concept. Values of 0.40–0.62 for six concepts indicate genuine directional agreement; moral valence (0.24–0.42 on the positive pole; feature 8529 at L19 shows strong anti-alignment at $\cos = -0.60$) is an outlier discussed in the text. *Bottom:* Mean $\pm$ 1σ normalised discrimination curves across all 7 concepts (left), and individual curves for moral valence and sentiment (centre, right) as illustrative examples of the per-concept curve structure.

The CAZ assembly events (L11, L13, L19) are not visible as peaks in either cumulative curve — they are visible in the *derivative*: the layers where the cumulative curve's slope is steepest. This is exactly what the CAZ deep-dive algorithm measures. The cumulative curve tells you where the concept is represented most fully; the derivative tells you where it was built.

Scope. This comparison covers 7 concepts against a 16,000-feature SAE. The SAE represents the full activation space; our eigenvectors are targeted probes for specific semantic axes. The question is not whether CAZ can replace SAE — it cannot, and is not designed to — but whether the two methods converge on the same structural signal for the concepts CAZ does cover. The positive evidence is (i) **direction alignment** at CAZ peak layers (cos 0.40–0.62 for six concepts, with the moral valence anti-alignment hypothesis above), and (ii) the **shared-peak finding** at L11, L13, and L19 — the SAE has no knowledge of our eigenvectors or peak-detection method, yet identifies the same layers as maximally polysemantic (28–30 shared features vs. ≤ 17 at other layers). These two pieces of evidence are what a third method converging on CAZ's structural findings would look like.

The curve correlation ($r = 0.986$) is a consistency check, not independent convergence evidence: both methods measure cumulative unnormalized separation and both grow monotonically as the residual stream accumulates concept-relevant information toward the final layer, so high correlation is the expected result given shared measurement basis — we report it to confirm that the two pipelines track the same underlying quantity, not to count it toward method convergence.

8.7 Limitations

Concept selection: Seven concepts were selected by the author based on definitional clarity and contrastive operationalizability (§2.2). No systematic survey of possible concepts was conducted, and no claim is made that these seven are representative of concept space generally. Any concept that is poorly served by contrastive pairs — those lacking clear antonyms, those requiring world knowledge rather than linguistic context, or those whose boundaries are culturally contingent — would be invisible to this methodology. The reported ordering ($\tau = 0.54$) and convergence patterns describe these seven concepts; extrapolation to a broader concept inventory is untested.

Concept coverage: Seven concepts capture 2% of organized activation structure. The dark matter question remains open.

Self-evaluation: All experiments, analysis, and interpretation in this paper were conducted by the same author who developed the CAZ framework. Independent replication by a group with no prior investment in the hypothesis would carry substantially more evidential weight. The dataset, code, and extraction outputs are released to enable exactly this.

Scale ceiling: Our largest model is 9B parameters (Gemma-2-9b, bf16, sharded across two L4s). Whether concept ordering and convergence hold at frontier scale (70B+) is untested. All experiments were conducted on commodity hardware (2× NVIDIA L4, 22 GiB each). Frontier-scale evaluation awaits access to larger compute.

Dataset provenance: Main-study concept datasets were generated by a single model (Claude Sonnet 4.5), introducing three specific risks — lexical confounding, register/formality gradient, and template convergence — detailed in §2.2. The ordering-level concern has been directly tested: the multi-generator consensus

replication (Supplementary §C; ~1,400 pairs per concept from 14 generators across 4 AI labs, rerun on 4 architecturally diverse models) preserves the concept ordering and produces only concept-specific depth refinements consistent with the framework's multimodal predictions. Pair-level audits (bag-of-words baseline, lexical overlap analysis, 20% human validation) remain as planned mitigations at the absolute-separation level; these do not reopen the ordering finding.

Dataset scale: All results in this paper — including the CAZ/SAE curve equivalence ($r = 0.986$) — are derived from 100 contrastive pairs per concept. This is sufficient to establish the reported effects, but a larger pair set would sharpen CAZ boundary estimates, reduce variance in depth measurements, and likely improve the direction-agreement numbers in §8.6. Diminishing returns are likely — the Fisher separation metric stabilizes quickly with paired data — but the current sample size has not been formally power-analyzed.

Credibility bimodality: The three distinct credibility behaviors (standard assembly, delayed assembly, and near-zero separation throughout) may reflect genuine architectural differences in how models encode epistemic assessment — or may indicate that our "credibility" contrastive pairs capture heterogeneous phenomena that different architectures parse differently. Credibility shows the highest within-concept variance (std 29.7 on separation at peak) of any concept. The bimodality is reported as a finding, not explained; the current data do not distinguish between these interpretations.

Suppression threshold: The 20% suppression threshold used to classify ablation results (§6) is a researcher degree of freedom. The gentle-to-non-CAZ enrichment ratio ranges from 1.2× at 5% to 10× at 50%; the corresponding gentle CAZ efficacy rate (% above threshold) ranges from approximately 98% at 5% to approximately 80% at 50%, with 93% at the 20% operating point reported throughout §6.1. The qualitative conclusion (gentle CAZes enriched relative to non-CAZ) holds across this range; the specific 93% figure should be read as a threshold-specific estimate, not a threshold-free property. No principled basis for choosing 20% over 15% or 25% is offered.

Statistical power for ordering: Kendall's τ computed over 7 concepts has limited discriminative power. With N = 7, the minimum τ distinguishable from zero at $p < 0.05$ is approximately 0.43 (two-tailed). Our per-model median of 0.54 exceeds this, but the margin is narrow — the test can confirm the ordering tendency exists while leaving substantial uncertainty about its magnitude. The pairwise mean of 0.24 falls below this per-model threshold, which is why we report the per-model median as the primary statistic. Extending to a larger concept set would sharpen this measurement.

MHA/GQA behavioral split: The cohort-level behavioral difference (§5, §8.3) is confounded with training era, dataset scale, and multiple co-varying architectural choices. This is discussed in detail in §8.3; the limitation is noted here for completeness.

Smoothing parameters: The velocity smoothing window ($k = \max(1, \lfloor L/24 \rfloor)$) and feature tracking threshold ($\cos > 0.5$) were not formally optimized. The smoothing window adapts to model depth; the cosine threshold was standardized at 0.5 across all 195 deep dive runs.

Zone-level ablation (GEM) scope: The zone-level causal validation reported in §6.5 uses a fixed ablation width parameter (width = 3 layers around the handoff point). For models with fewer than 12 layers, this width represents a substantial fraction of total depth and may overestimate handoff suppression by interfering with nearby non-target processing. The pilot results in §6.5 are from models with 24–48 layers where this concern does not apply, but the parameter has not been formally sensitivity-tested. Full GEM methodology, width-sensitivity analysis, and results across the 34-model sweep are described in a companion paper [Henry, 2026f, *in preparation*]; §6.5 reports only the headline findings relevant to the zone-level hypothesis test.

Detection null model scope: The permutation null model (§4.1) was run on 5 of 26 models. The results are consistent across all 5 architectures tested, but formal null-model coverage for the remaining 21 models is not provided. The null model tests label permutation (random pos/neg assignment within existing pairs). The stronger null — random contrast directions in activation space at the same CAZ peak layer — is addressed by the direction specificity control in §6.1 (median 282× specificity ratio, 97.8% of pairs concept direction exceeds all random seeds).

Linearity assumption: All metrics assume that concept information is linearly separable in the residual

stream. Nonlinear concept encodings would be invisible to our probes.

Multiple comparisons: No formal multiple-comparison correction is applied across the 26 × 7 × L test grid. Most reported statistics are aggregated (cohort means, median τ, permutation z-scores), which partially mitigates the concern — the 3.67× ablation enrichment and the $\tau = 0.54$ ordering tendency are each single tests over pooled data. However, concept-level breakdowns (e.g., the 45% divergence rate across 80 cases in §6.5, per-concept credibility split rates) involve implicit multiple testing that is not corrected. The relay-feature permutation $p = 0.001$ (§7) is particularly exposed: the candidate-discovery pipeline applies multiple filtering rules before reaching the permutation test, and the p-value is not Bonferroni-adjusted for the number of rules tried. If five pipeline stages were applied, the family-wise threshold becomes $p < 0.0002$ and the observed $p = 0.001$ no longer clears it; if ten stages were applied, it does not clear at any standard threshold. Because the number of discovery-rule comparisons is not systematically documented in this paper, the relay p-value should be treated as a lower bound on the true adjusted p. This is an additional reason — beyond the dominant-variance and circularity concerns noted in §7 — that the relay feature count is provisional and should not be cited as a confirmed result.

Behavioral validation gap (partially addressed): All causal claims in §6 are operationalized within activation geometry. Ablation suppresses Fisher separation; patching recovers it. The behavioral pilot in §6.8 (5 models × 7 concepts, logit-difference suppression) confirms direction-specific suppression at CAZ peak layers (random ablation 0), but the single-layer design has the same limitation as all geometric ablation experiments in this paper: concept directions are distributed across many layers, so any single layer carries partial signal and single-layer peak ablation does not dramatically outperform a midpoint control. The appropriate behavioral test is zone-level — ablating across the full CAZ window rather than at a single peak layer — analogous to the GEM zone-level intervention [Henry, 2026f] that reveals far stronger effects in the geometric domain. Zone-level behavioral suppression, along with task-accuracy measurement across a broader model set, remains as planned follow-up work.

Patching baseline gap: Activation patching experiments (§6.4) were performed at identified CAZ peaks. No systematic comparison against patching at non-CAZ layers was conducted. The "sufficient" characterization is therefore relative to the ablation results (which include a non-CAZ baseline via the 3.67× enrichment), not to a patching-specific control. Future work should establish patching effectiveness at random or uniformly-sampled non-CAZ layers.

Single-layer ablation gap: Every ablation experiment removes the concept direction at a single layer. The CAZ framework hypothesizes that concepts assemble across a window of layers, yet all interventions are point tests. Single-layer ablation may understate concept presence in architectures where the concept direction rotates across the assembly window. A proper zone-level intervention would capture the concept's trajectory across the full CAZ window, not just its instantaneous direction at one depth. The post-CAZ single-layer approach performs well because it catches the concept after assembly is complete; it does not test the assembly process itself.

KL divergence summary: KL divergence from the unablated next-token distribution was computed for all ablation sweeps but not reported in §6. CAZ peaks are not privileged low-damage intervention points: across all cohorts, KL at CAZ peaks is not significantly lower than at non-CAZ layers, and is higher in GQA and Gemma-2 architectures ($p = 0.07$ and $p = 0.016$ respectively). However, the post-CAZ efficiency advantage does extend to prediction-level disruption: post-CAZ ablation achieves a median efficiency of 47.0 (separation reduction per unit KL) versus 32.2 at the CAZ peak and 15.7 pre-CAZ in MHA models. This confirms that post-CAZ layers are the most targeted intervention point by both geometric and prediction-level metrics.

Spectral method limitations: Three structural limitations of our eigenvalue-based feature detection deserve explicit acknowledgment. First, SVD is bounded by the hidden dimension d — it cannot resolve features in superposition, so a single detected feature may be a macro-cluster of finer-grained features that a sparse autoencoder could separate. Second, the orthogonality constraint of eigenvector decomposition may slice the model's native geometry at artificial angles, imposing structure that reflects the method rather than the model. Third, the 94% dark matter fraction is relative to our specific input distribution — different calibration texts probing different capabilities would discover different features. Our flashlight illuminates one patch of the warehouse; the warehouse is larger than the patch.

9. Conclusion

The Concept Allocation Zone framework survives systematic evaluation across 26 base transformer models from 8 architecture families. Three findings are robust: a statistically detectable concept ordering tendency (median per-model $\tau = 0.54$, z = 11.5, $p < 0.001$; 87% of models positively correlated), causal reality of gentle CAZes (half the causal structure was previously invisible, with 4.5× enrichment over non-CAZ baseline), and confirmed one-way information flow through the allocation pipeline.

Two predictions are not supported (P4, P6) and one is partially supported with a falsified peak-level component (P1). P4 (concept handoffs) was invalidated by multimodal allocation. P6 (lexical vs. compositional identity) failed at p = 0.82. P1's peak-level failure — within-CAZ is the modal optimal class (56.9%) but 32.4% of optima are post-CAZ, and the peak itself is rarely optimal — led to the discovery of a descriptive cohort difference: the MHA+GELU group (2019–2022 models) encodes concepts redundantly; the GQA+SwiGLU group (2023–2025 models) encodes them sparsely; Gemma-2 (alternating attention, n=2) localises at its final global layer. The cohort effect is an empirical regularity; the mechanistic attribution remains open. Two positive controls narrow the space: the Pythia RoPE experiment rules out position encoding specifically, and the patching-asymmetry argument constrains a particular signal-distribution variant of the training-scale account. Neither establishes architecture as the cause — they rule out two narrow alternatives, not the broader era/scale family. GQA+SwiGLU remains the most tractable architectural hypothesis, but training data scale, curriculum, and dataset composition all co-vary across cohorts and the decisive experiment (training-scale ablation within a fixed architecture) has not been run. P4's failure exposed the multimodal allocation structure and velocity X-crossings that the single-peak framework had obscured. P6's failure narrows the design space for future tests of the shallow/deep distinction.

Concepts assemble multimodally, with depth-separated sub-representations (shallow vs deep peaks, within-model cos 0.15–0.43) that independently converge across architectures. At shallow depths, semantically related concepts share dominant directions before diverging — consistent with either shared computational primitives or incomplete separation (§4.5). At depth, relational concepts merge while epistemic and affective concepts diverge — patterns that hold with zero exceptions across 20 same-dimension model pairs.

Ninety-eight percent of organized activation structure remains unlabeled by our seven concept probes. This dark matter is not noise — it is geometrically coherent and persistent across layers. Twelve candidate relay features appear in the dark matter pool (§7), but dominant-variance and circularity concerns mean the count is provisional; full validation is Paper 5 [Henry, 2026e]. Characterizing this structure is the critical next step.

A natural extension is fitting parametric functions to the per-concept separation curve $S(\ell)$ — treating each model × concept profile as a parameter vector rather than a raw trace. Deviations from the expected functional form would serve as diagnostics: flagging dataset quality issues, detection artifacts, or anomalous model behavior. Combined with CKA-based layer similarity fingerprints, this would provide a compact, quantitative description of how concept encoding varies across architectures — and a principled basis for targeting ablation interventions at fitted peaks rather than raw velocity extrema.

All data, code, and extraction scripts are publicly available at github.com/jamesrahenry/caz_scaling. Replication on additional architecture families, extended concept sets, or frontier-scale models is invited; correspondence to jamesrahenry@henrynet.ca.

Supplementary Materials

Three supporting documents accompany this preprint in the companion file `supplementary.md`:

- **§A · Technical Notes** — float64 metrics, OPT embedding projection handling, Marchenko–Pastur threshold definition.

- **§B · Per-Model Behavioral Profiles** — complete CAZ / score / %Gentle / black-hole / persistent-feature tables for all 35 models (26 base + 9 instruct), plus per-family commentary and cross-family observations.
- **§C · Multi-Generator Consensus Replication** — Peak-depth comparison between the 100-pair single-generator dataset and the ~1,400-pair 14-generator consensus dataset for Pythia-1.4b, GPT-2-XL, Qwen-2.5-3B, and Llama-3.2-3B.

Raw extraction outputs, ablation JSONs, patching JSONs, and the plotting code for every figure in this paper are released at the repository linked above.

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